Provenance and Causal Awareness in Personalized Advice

Private Values and Portable Beliefs

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Abstract

Personalized AI advice combines evidence about an objective state with adjustments for a user's private values. Reuse after the original context disappears can move a legitimate value adjustment into the factual category, which creates *preference-belief laundering*. This paper develops a two-date platform-user commitment game. The platform chooses persistent provenance before the user selects value disclosure and later chooses delegation plus costly audit. A global curvature condition yields a unique user response. Personalized advice improves current action quality for every type under the benchmark implementation assumptions. With sufficiently costly audit, later factual loss exceeds the no-AI benchmark for an exact interval of intermediate reports. A platform that internalizes only a fraction of continuation losses selects less provenance than a planner under the stated convexity conditions. Correct causal awareness removes the distortion. The framework links personalized advice to bounded representation through an endogenous choice over source persistence.

Keywords: awareness; belief; decision theory; artificial intelligence; provenance
JEL codes: C72, D80, D81, D83

1 Introduction

Personalized AI advice combines two economic functions within one recommendation. It conveys evidence about an objective state and adapts an action to user-specific circumstances. Strategic communication studies how incentives constrain informative messages (Crawford and Sobel, 1982). Bayesian persuasion places the information structure under commitment (Kamenica and Gentzkow, 2011). Recent GEB work on human-AI integration studies a decision maker plus an AI agent that hold separate informative signals about an unknown state (Lagziel and Tsodikovich, 2025). Personalized advice adds a value channel because user-specific circumstances can legitimately change the recommended action even when they convey no information about the state.

Portability makes that distinction consequential beyond the original interaction. Users can save AI responses for later use. Platforms can compress old conversations into shorter records. A recommendation may therefore survive after the language that

explained part of its origin has disappeared. Experiments on personalized pricing show that users alter reported information when they understand how those inputs affect algorithmic outcomes (Bó et al., 2024). Models of selective memory show that uneven retention can change later beliefs when remembered experiences receive full inferential weight (Fudenberg et al., 2024). The present problem concerns the source classification attached to surviving personalized content.

Source classification matters because factual evidence and private values play different economic roles. Let θ denote an objective state. Let κ denote a private value that changes the action suited to the user. The benchmark gives κ no direct information about θ. A closely related separation appears in social learning when observed actions depend on beliefs about the state as well as idiosyncratic types; small errors about the type distribution can then generate large learning failures (Frick et al., 2020). Personalized AI creates a different route to the same broad concern because value-dependent content can travel inside a recommendation. Awareness theory supplies the language for a further possibility: the represented model may omit the source proposition required to interpret retained content (Fagin and Halpern, 1987).

This paper studies persistence of that source proposition. *Provenance* records why a component entered the recommendation and which source generated it. *Persistent provenance* denotes the share of source information that remains attached after the original interaction has been compressed or saved. Causal awareness of value origin requires the represented model to retain the proposition that part of the recommendation came from the user's reported value. Loss of that proposition moves residual value-derived content into the factual category. The resulting source-classification error receives the term *preference-belief laundering*.

A portfolio recommendation provides a concrete example. Consider an investor asking an AI system how much of a risky asset to hold. The system has evidence about expected return. The investor reports a liquidity need, such as a near-term need for cash. Expected-return evidence describes the asset, while the liquidity need changes the portfolio suited to the investor. A larger liquidity need can rationally reduce the recommended risky position without changing the system's estimate of expected return. Date 1 represents that original portfolio decision. Date 2 represents later fact-only reuse, which can involve reconstructing what the earlier evidence implied about expected return. If the saved record retains the conservative portfolio recommendation

after losing the liquidity rationale, the conservative adjustment can enter the later inference as evidence of low expected return. The original recommendation can therefore improve the portfolio while degrading a later factual belief.

The mechanism can be summarized by the chain

$$\begin{aligned}\kappa &\longrightarrow \text{value disclosure in the prompt}\\ &\longrightarrow \text{personalized AI recommendation}\\ &\longrightarrow \text{portable factual belief.}\end{aligned}$$

Value disclosure changes the personalized recommendation. Persistent provenance plus private audit determine how much value-derived content retains its source classification when the recommendation becomes portable. Residual unlabeled content enters the later factual category. The chain links a legitimate input to the original action with a potential error in a different decision after source context has disappeared.

The formal environment turns this problem into a two-date platform-user commitment game. The platform commits to persistent provenance before value disclosure. The user observes a private factual signal plus a private value and reports that value to the system. The AI response contains factual information together with an intended value adjustment. Full date-1 context lets the user separate the value adjustment from factual inference. The user also chooses delegation of value implementation and costly audit of unlabeled personalized content. Date 2 retains a compressed record under the benchmark memory assumption, while the historical report and its claim-level mapping have disappeared. The AI response technology remains fixed across dates. Distortion comes from the source language available to the later represented model.

The user problem yields a sharp divergence between current action quality and later factual quality. Conditional on a report, delegation plus audit admit closed-form solutions. Substitution reduces disclosure to one dimension, and a global curvature condition gives a unique report for every private type. Personalization has a genuine current benefit because human value implementation and AI value implementation both contain error. The later effect follows the audit response. Small reports leave little value-derived content. Large reports induce intensive audit under the benchmark technology. Intermediate reports can therefore leave the largest residual contamination. Sufficiently costly audit produces an exact interval of intermediate reports for which date-2 factual loss exceeds the no-AI benchmark. This type-level harm can coexist with

lower population-average factual loss because the AI also contributes clean information. Full intertemporal welfare additionally includes reporting costs plus audit expenditure.

Persistent provenance then enters the platform design problem. A planner internalizes the full continuation burden created by later factual distortion plus private audit. The platform internalizes a fraction χ. Stronger provenance preserves the causal reason behind personalized content and changes user disclosure through anticipated continuation risk. Under the theorem's convexity conditions, partial internalization leads the platform to choose less provenance. A complete causal record removes the laundering channel and restores the associated Blackwell comparison (Blackwell, 1953). Imperfect labels can suppress valid factual information, so provenance design requires accurate source classification.

The contribution concerns an intertemporal source problem that lies between advice design and bounded representation. Advice models determine how a recommendation maps information into a current action or belief. Representation models determine how observed content receives statistical meaning. The present framework connects those stages by allowing personalized content to survive after its source proposition has disappeared. Platform provenance plus private audit govern persistence of that proposition. Anticipated source loss feeds back into disclosure before the recommendation has been generated. A primitive capacity constraint also links factual precision to intended personalization, which keeps capability comparisons on feasible technology paths.

Section 2 defines the two decision problems and constructs the portable record. Section 3 solves the user problem and separates current action quality from later factual loss. Section 4 studies the platform commitment problem. Section 5 gives the causal-awareness boundary and quantitative illustrations. Proofs plus verification details appear in the appendices.

1.1 From information transmission to causal awareness

The communication literature establishes how the process generating advice shapes informational content. Preference conflict limits informative messages in strategic communication (Crawford and Sobel, 1982). Bayesian persuasion places the information structure under commitment (Kamenica and Gentzkow, 2011), while Bayes correlated equilibrium extends comparisons across information structures to games (Bergemann

and Morris, 2016). Advisor models bring generation rules closer to the present setting. Conflict disclosure changes informativeness under delegation (Gesche, 2021). Hidden motives make transparency about advisor incentives a policy variable (Onuchic, 2025). These contributions motivate attention to the process behind a recommendation. The present paper shifts attention to whether information about that process survives after the recommendation travels to another decision.

Human-AI advice research provides a closer benchmark for the date-1 information structure. Lagziel and Tsodikovich study rational integration when the decision maker plus the AI each hold informative private signals about an unknown state (Lagziel and Tsodikovich, 2025). Experimental work shows that reliance on algorithmic advice can fall after observed algorithmic errors (Dietvorst et al., 2015). Other experiments document algorithm appreciation in direct comparisons with human advice (Logg et al., 2019). Task characteristics shift algorithmic reliance (Castelo et al., 2019). Incentive design and human framing can also change uptake of AI-generated advice (Greiner et al., 2026). These papers study whether decision makers use algorithmic information. The present model adds a separate source-classification problem after personalized advice has been reused.

Platform models make privately held information part of the design environment. Reputation-based persuasion platforms condition policy on information supplied by users (Arieli et al., 2024). Platforms may restrict information acquisition when users hold biased beliefs (Filiz-Ozbay and Liu, 2026). Personalized-pricing experiments show strategic manipulation of reported characteristics when consumers understand that a statistical model maps those responses into prices (Bó et al., 2024). Ichihashi and Smolin place privately held seller information inside an information-design problem in which a designer supplies additional information to a seller already informed about buyer value (Ichihashi and Smolin, 2025). Their analysis concerns screening and welfare under information provision to an informed seller. Related platform models also show that incomplete recorded histories can matter for subsequent strategic outcomes in repeated credence-goods markets (Gradwohl and Jelnov, 2024). The present mechanism adds a different persistence margin: value-derived content survives while its causal source category can disappear.

Bounded-representation models explain why economically relevant distinctions may drop out of the model used for inference. Coarse thinking merges categories preserved by a richer representation (Mullainathan et al., 2008). Selective attention makes

learning depend on which variables enter the represented model (Schwartzstein, 2014). Sparse attention provides a choice-based foundation for simplified internal models (Gabaix, 2014). Local thinking links selective recall to distorted judgment (Gennaioli and Shleifer, 2010). Analogy-based expectations group distinct situations into common classes (Jehiel, 2005). Cursed equilibrium places coarse inference inside strategic interaction (Eyster and Rabin, 2005), while Berk-Nash equilibrium formalizes behavior under misspecified models (Esponda and Pouzo, 2016). A recent review organizes this broader literature around learning with incorrect models of the information environment (Bohren and Hauser, 2025). Preference-belief laundering uses a particular omitted distinction: the source category separating value adjustment from factual evidence.

A second set of models clarifies why source or type classification can matter even when the underlying content remains observed. Persuasion bias arises when repeated information receives excessive weight (DeMarzo et al., 2003). Correlation neglect changes inference when dependence across observations receives the wrong representation (Eyster and Rabin, 2010; Levy and Razin, 2015; Enke and Zimmermann, 2019). Social learning becomes fragile when observed actions depend on idiosyncratic types and observers slightly misperceive the type distribution (Frick et al., 2020). Spiegler studies a procedural setting in which users fail to distinguish raw data from imputed data, producing distorted correlation and causal structure (Spiegler, 2017). Noisy proxies create a related problem when an imperfect measurement receives excessive factual status (Clyde, 2025). The present setting places a user value inside a recommendation and studies later factual reuse after the value origin has been lost.

Memory models provide the intertemporal link from an original interaction to later inference. Limited memory can generate systematic information-processing biases when retained summaries substitute for the full history (Mullainathan, 2002; Wilson, 2014). Selective-memory equilibrium studies agents who remember some experiences more often and update as if remembered experiences exhaust the realized history (Fudenberg et al., 2024). GEB work on memorable consumption gives recollection a direct role in intertemporal welfare (Minardi and Savochkin, 2024). The present model uses memory for a narrower informational purpose. Compression keeps part of the recommendation while changing which source propositions remain available at reuse.

Awareness theory provides the formal language for that missing source proposition.

Explicit awareness sets allow the represented language to omit a relevant proposition (Fagin and Halpern, 1987). Alternative semantics develop unawareness through different state representations (Halpern, 2001). Generalized state spaces extend interactive unawareness (Heifetz et al., 2008). Information structures with unawareness separate factual content from awareness content (Li, 2009). Ambiguous interpretation can generate different beliefs from shared observations (Halpern and Kets, 2015). The economic step in this paper makes survival of the source proposition endogenous. Persistent provenance plus private audit determine the causal language available at date 2, while anticipated exposure changes date-1 disclosure.

2 Sequential Advice Game

2.1 Two tasks and factual information

The two dates represent different uses of the same underlying factual state. Date 1 captures the original personalized decision, where the user's private value affects the appropriate action. Date 2 captures later fact-only use, where the historical value adjustment no longer enters the target. The factual environment combines a common Gaussian state with a private human signal. Let the objective state satisfy

$$\theta \sim \mathcal{N}(0, \tau_0^{-1}), \qquad \tau_0 > 0.$$

The parameter τ_0 denotes prior precision. The user observes

$$x = \theta + \varepsilon_H, \qquad \varepsilon_H \sim \mathcal{N}(0, \tau_H^{-1}).$$

The parameter τ_H denotes precision of the human factual signal. Define

$$T = \tau_0 + \tau_H, \qquad \mu_x = \mathbb{E}[\theta \mid x].$$

Thus T denotes pre-AI factual precision, with μ_x denoting the corresponding posterior mean. The user has a private date-1 value κ drawn from a symmetric continuous density with full support and finite variance σ_κ^2. The baseline imposes $\theta \perp \kappa$. Mutual independence applies to the factual noises ε_H and ε_q, with both independent of (θ, κ). The primitive implementation below also imposes mutual independence between ε_I

and ζ, with both independent of $(\theta, \kappa, \varepsilon_H)$. The induced ε_q inherits the required independence property.

The independence assumption gives κ a sharp economic interpretation in the benchmark. The value shifts the action suited to the user at date 1 while carrying no information about the factual state by itself. Any date-2 effect of κ on beliefs must therefore pass through the recommendation channel and the representation of the portable record. Recent GEB work also studies human-AI integration with separate informative signals about a common state (Lagziel and Tsodikovich, 2025). The present environment adds a private value that changes the date-1 target and later creates the source-classification margin.

The private value enters the date-1 target because the original decision uses personalization. The ideal personalized action satisfies

$$a_1^* = \theta + \omega\kappa, \qquad \omega > 0. \tag{1}$$

At date 2, the user faces a fact-only action:

$$a_2^* = \theta. \tag{2}$$

The parameter ω denotes the ideal loading on the private value. In the portfolio example, θ captures expected return, while $\omega\kappa$ captures the portfolio tilt justified by the investor's liquidity need. Date 1 asks which portfolio suits that investor, so the target uses both components. Date 2 asks for the factual state alone, which leaves the earlier liquidity adjustment outside the target.

2.2 Personalized advice and technological feasibility

The AI recommendation has a factual component and a personalization component. The factual component comes from an independent signal about θ. The AI observes

$$z = \theta + \varepsilon_I, \qquad \varepsilon_I \sim \mathcal{N}(0, \tau_I^{-1}).$$

The parameter τ_I denotes precision of the primitive AI factual signal. The user sends a value report v. A normalized composite recommendation satisfies

$$y = \theta + \varepsilon_q + Pv, \qquad \varepsilon_q \sim \mathcal{N}(0, \tau^{-1}). \tag{3}$$

The coefficient $P \geq 0$ measures intended personalization, while $\tau > 0$ measures effective factual precision. Equation (3) separates factual information $\theta + \varepsilon_q$ from the value adjustment Pv generated by the report. Later reuse makes the source of Pv economically relevant.

Similar-looking recommendations can arise from different causes. A change in factual evidence can move the recommendation. Adaptation to a different reported value can produce the same movement. Full date-1 context keeps the causes distinguishable, while compression can remove that distinction from the portable record.

A simple primitive implementation clarifies feasibility. The coefficient α loads the reported value. The coefficient β loads the factual signal. Let the unnormalized response be

$$Y = \beta z + \alpha v + \zeta, \qquad \zeta \sim \mathcal{N}(0, \sigma_\zeta^2), \qquad \sigma_\zeta^2 > 0.$$

The parameter σ_ζ^2 denotes primitive response-noise variance. The capacity constraint takes the form

$$\alpha^2 + \beta^2 \leq 1, \qquad \alpha \geq 0, \qquad \beta > 0.$$

Then

$$P = \frac{\alpha}{\beta}, \qquad \tau = \left[\tau_I^{-1} + \frac{\sigma_\zeta^2}{\beta^2}\right]^{-1}. \tag{4}$$

Solving for β yields a precision-personalization frontier.

Lemma 1 (A primitive feasibility frontier)**.** *Under* (4)*, every feasible design satisfies*

$$P^2 \leq \overline{P}(\tau)^2 = \frac{\tau_I - \tau - \sigma_\zeta^2 \tau \tau_I}{\sigma_\zeta^2 \tau \tau_I}, \tag{5}$$

for

$$0 < \tau \leq \left(\tau_I^{-1} + \sigma_\zeta^2\right)^{-1}.$$

Feasibility of action-correct personalization $P = \omega$ requires

$$\tau \leq \left[\tau_I^{-1} + \sigma_\zeta^2(1 + \omega^2)\right]^{-1}. \tag{6}$$

The main results use (P, τ) and apply to any technology that can implement them. Lemma 1 embeds these reduced-form objects in an explicit primitive capacity constraint. The frontier gives the reduced-form parameters a technological interpretation: with the primitive signal environment held fixed, greater effective factual precision and stronger intended personalization cannot be varied as completely independent design knobs. This restriction keeps the later capability comparisons on feasible technology paths.

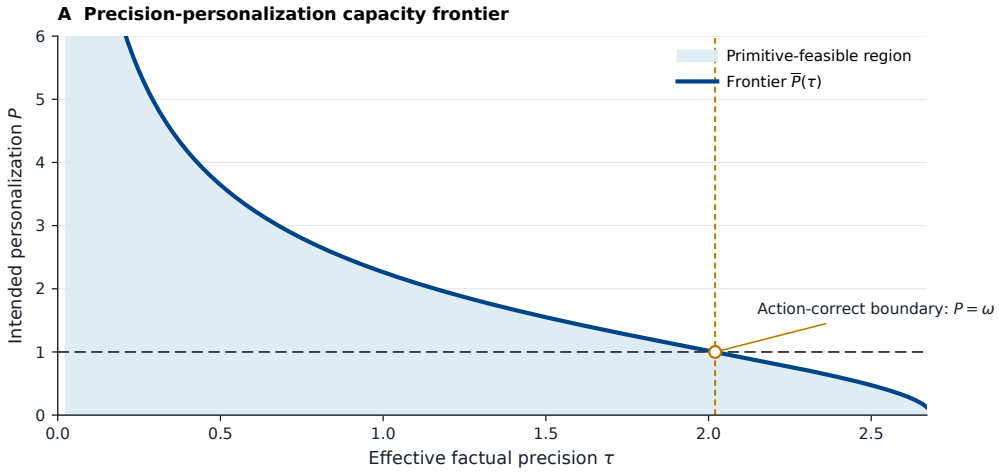


Figure 1: A primitive capacity technology links factual precision to intended personalization. The action-correct loading remains feasible only below the action-correct feasibility limit. The illustration sets $\tau_I = 4$ and $\sigma_\zeta = 0.35$; $\omega = 1$.

2.3 Source-aware use and implementation

Date-1 inference uses the original prompt together with the response mapping, which preserves the source of Pv. The user therefore removes that value adjustment when extracting factual information about θ. The resulting source-aware posterior satisfies

$$\widehat{\theta}^R = \mu_x + w[(y - Pv) - \mu_x], \qquad V = \frac{1}{T + \tau}, \qquad w = \frac{\tau}{T + \tau}. \tag{7}$$

The quantity V denotes clean posterior variance after combining the factual signals. The weight w measures posterior reliance on the AI factual signal. The same coefficient governs the date-2 effect of residual value-derived content, so stronger reliance on the AI factual channel also amplifies content that later enters that channel through misclassification.

Delegation governs the implementation of the private value at date 1. The user has a human value-translation technology, and the AI supplies a second imperfect translation. The realized human and AI adjustments take the form

$$u_H = \omega\kappa + \eta_H, \qquad u_A = Pv + \eta_A,$$

where

$$\eta_H \sim \mathcal{N}(0, A_H), \qquad \eta_A \sim \mathcal{N}(0, B_A), \qquad A_H, B_A > 0.$$

The parameters A_H and B_A denote implementation-error variances for the human and AI technologies. The delegation share ρ places weight on the AI value adjustment, with $1 - \rho$ placed on the human adjustment. The implementation errors have zero mutual covariance and remain independent of every factual shock plus κ. Realization occurs after the delegation choice. The user chooses $\rho \in [0, 1]$ before realization of (η_H, η_A) and takes

$$a_1 = \widehat{\theta}^R + (1 - \rho)u_H + \rho u_A. \tag{8}$$

The parameter ρ represents delegation of value implementation. The factual forecast has already entered through $\widehat{\theta}^R$. Behavioral evidence shows that reliance on algorithmic advice can change after observed errors (Dietvorst et al., 2015). Task characteristics also shift reliance (Castelo et al., 2019), while direct comparisons sometimes produce algorithm appreciation (Logg et al., 2019). Incentive design and human framing can further change advice uptake (Greiner et al., 2026). The model abstracts from those behavioral determinants and lets ρ optimize the noisy implementation technologies stated above.

2.4 Portable records and persistent provenance

Persistent provenance records why personalized content remains in a compressed recommendation. A source label can identify a claim as partly generated from the user's report, which allows removal of that component from later fact-only use. Natural-language responses can distribute the loading Pv across multiple claims. The platform attaches persistent source labels to share $s \in [0, 1]$ of the value-derived content. Before compression, the user can audit share $\ell \in [0, 1]$ of the remaining unlabeled content

and separate it from material retained as factual evidence. The classified share equals

$$\lambda(s, \ell) = s + (1 - s)\ell.$$

Future fact-only use excludes classified value content. The remaining portable AI record satisfies

$$r_2 = y - \lambda(s, \ell)Pv = \theta + \varepsilon_q + (1 - s)(1 - \ell)Pv. \tag{9}$$

Definition 1 (Causal awareness)**.** *Causal awareness of value origin holds when the represented model contains the proposition that part of an observed recommendation came from a value report. The associated likelihood can then separate that component from factual evidence. Causal unawareness removes this proposition from the represented model, so residual unlabeled content enters the available factual category.*

At date 1, the report and response mapping preserve causal awareness of value origin. Persistent provenance carries the source proposition forward for a labeled share. Private audit carries it forward for an additional share. In the portfolio example, these mechanisms preserve the liquidity rationale attached to part of the recommendation and prevent that component from entering expected-return inference.

Assumption 1 (Portable memory and causal unawareness)**.** *At date 2, the retained information takes the form*

$$\mathcal{I}_2^N = \sigma(x, r_2, \tau). \tag{10}$$

The date-2 information set omits the historical type κ plus report v. The record also omits the original prompt and raw response y. Compression removes the claim-level value mapping. The represented model therefore places every remaining unlabeled component of r_2 in the factual-signal category and uses the likelihood $r_2 = \theta + \varepsilon_q$.

The user chooses audit before source context disappears and therefore anticipates the later classification error. The date-1 objective evaluates true expected date-2 loss under the represented decision rule and trades that loss against cleaning cost. After compression, the date-2 self applies the represented posterior to the surviving information. The term $(1 - s)(1 - \ell)Pv$ then remains in the record after its value origin has disappeared from the represented model.

The date-2 posterior inherits any residual value loading. Under Assumption 1, the

represented portable posterior satisfies

$$\widehat{\theta}^N = \mu_x + w(r_2 - \mu_x) = \widehat{\theta}^R + w(1-s)(1-\ell)Pv. \tag{11}$$

Equation (11) makes the source-loss channel explicit: residual value loading enters the date-2 posterior through the factual-signal weight w. A record that retains (v, P) allows the value loading to be removed directly. Lemma 2 gives the corresponding causal-awareness result.

The model nests several memory environments within one state space. An answer-only record corresponds to little persistent classification. The portable record then carries most of the unaudited value loading. A persistent value label corresponds to an interior s and removes the labeled share before factual reuse. Cheaper marking or deletion raises the user's optimal audit share. Under a complete causal record, the date-2 self observes the relevant report and loading. The self then removes the value component. These memory environments map persistent provenance into the causal source language available at reuse. Models of bounded memory show how retained summaries can change later information processing (Mullainathan, 2002; Wilson, 2014). Selective-memory equilibrium provides another route from selective retention to distorted updating (Fudenberg et al., 2024). Related GEB work assigns recollection a direct intertemporal welfare role (Minardi and Savochkin, 2024). Spiegler's raw-versus-imputed distinction offers a closer analogy to source classification because users process differently generated data as one practical belief (Spiegler, 2017).

2.5 Timing

The timing separates the original personalized decision from later reuse. The game unfolds in six steps. Common knowledge covers the response technology (P, τ) plus the type distribution. The platform chooses one public provenance policy for all users and observes the type distribution without observing any user's realized (x, κ). The sequence follows:

1. The platform commits to persistent provenance s before prompt disclosure.
2. The user observes (s, x, κ) and chooses the value report v.
3. The AI mechanically draws y from the response kernel in (3).
4. With the full context available, the user forms $\widehat{\theta}^R$. The user then chooses delegation ρ and audit ℓ. Delegation precedes the implementation errors.

5. Realizations of (η_H, η_A) occur. The date-1 action follows. Storage of the portable record r_2 occurs next; the original context subsequently disappears under Assumption 1.
6. The date-2 fact-only task occurs. The date-2 self acts on $\mathcal{I}_2^N$ using (11).

The timing leaves marginal policies independent of the realized factual signal. The date-1 self anticipates the imposed date-2 rule. A single expected intertemporal objective governs all user choices. Quadratic-normal location invariance makes the marginal policies independent of μ_x. The signal x shifts the level of both posterior actions while leaving these policies unchanged.

Definition 2 (Behavioral commitment outcome)**.** *Fix (P, τ) and a reference type distribution. A behavioral commitment outcome specifies a common public choice s plus a report policy $v(s, x, \kappa)$. Continuation policies take the forms $\rho(s, x, \kappa, v, y)$ and $\ell(s, x, \kappa, v, y)$. Equation* (11) *governs date-2 action and minimizes represented loss. The continuation policies minimize sophisticated date-1 continuation loss. The report policy minimizes user loss in anticipation of those policies. The platform minimizes* (64) *under the induced type distribution. Backward induction under Assumption 1 delivers the solution.*

The quadratic-normal structure further collapses the continuation policies. These policies reduce to $v_s^*(\kappa)$ plus $\rho^*(v, \kappa)$, with audit given by $\ell_s^*(v)$. Date-2 action follows $\widehat{\theta}^N$.

3 User Subgame and Intertemporal Welfare

3.1 The user's problem

The user's objective formalizes the tradeoff described in the introduction. A more informative value report can improve the current personalized action while placing more value-derived content into the portable record. Costly information acquisition provides a broad analogue for the audit margin because attention responds to the value of additional information (Caplin and Dean, 2015). Here the acquired information concerns causal provenance before reuse. The parameter $\delta_F \in (0, 1]$ weights the date-2 task. The coefficient $c_v > 0$ keeps the report close to the user's actual value. The coefficient $c_p \geq 0$ captures the direct cost of revealing that value. The coefficient $c_\ell > 0$

makes source audit costly. A report that differs from κ incurs fidelity cost $c_v(v-\kappa)^2/2$. Reporting values has privacy or disclosure cost $c_p v^2/2$. Audit costs $c_\ell \ell^2/2$. Taking expectations before realization of implementation errors, the user minimizes

$$\begin{aligned}\mathcal{C}(v,\rho,\ell;\kappa) = (1+\delta_F)V + \rho^2\left[B_A + (Pv-\omega\kappa)^2\right] + (1-\rho)^2 A_H \\ + \delta_F w^2(1-s)^2P^2v^2(1-\ell)^2 + \frac{c_\ell}{2}\ell^2 + \frac{c_v}{2}(v-\kappa)^2 + \frac{c_p}{2}v^2.\end{aligned} \tag{12}$$

The objective separates clean factual risk from behavioral costs. The constant term $(1+\delta_F)V$ captures clean factual variance across both tasks. Current implementation loss appears on the first line. The second line collects future contamination and reporting-related costs. Audit expenditure appears there as well. Equation (12) puts the central tradeoff in one place. More faithful reporting can improve current personalization, while a larger value component can create more exposure at reuse. Audit reduces that exposure at a direct cost.

The continuation problem can be compressed into a small set of reduced-form objects. M summarizes the AI implementation loss associated with a given report. D summarizes the strength of future contamination before audit. The function f gives current implementation loss after optimizing delegation. Define

$$M(v,\kappa) = B_A + (Pv-\omega\kappa)^2, \tag{13}$$

$$D = \delta_F w^2(1-s)^2P^2, \tag{14}$$

$$f(t) = \frac{A_H(B_A+t^2)}{A_H+B_A+t^2}. \tag{15}$$

These objects keep the current implementation margin and the future source-loss margin visible in the reduced report problem.

3.2 Optimal continuation response

For a fixed report, delegation determines the share of the current value adjustment implemented through the AI. Audit determines the amount of residual value-derived

content cleaned before reuse. Both choices have closed forms:

$$\rho^*(v,\kappa) = \frac{A_H}{A_H + M(v,\kappa)}, \tag{16}$$

$$\ell^*(v) = \frac{2Dv^2}{c_\ell + 2Dv^2}. \tag{17}$$

The delegation rule gives more weight to AI when human implementation carries more noise. Greater AI noise or poorer value matching reduces that weight. The audit rule rises with the size of the report because larger value adjustments create more potential contamination. After substituting these choices, the report solves

$$\min_v \; R(v;\kappa,D) = f(Pv - \omega\kappa) + \frac{c_\ell D v^2}{c_\ell + 2Dv^2} + \frac{c_v}{2}(v-\kappa)^2 + \frac{c_p}{2}v^2. \tag{18}$$

Uniqueness requires the direct curvature from reporting costs to dominate the worst local nonconvexity created by implementation and audit. A single curvature buffer summarizes that sufficient condition. Define

$$\underline{c} = c_v + c_p - \frac{A_H^2 P^2}{2(A_H + B_A)^2}. \tag{19}$$

Theorem 1 (Unified user response)**.** *Suppose* $P > 0$ *and*

$$\boxed{\underline{c} > \frac{D}{2}.} \tag{20}$$

Then the user problem has a unique optimal response (v^*, ρ^*, ℓ^*) *for every* κ*. Equations* (16) *and* (17) *give delegation plus audit. The unique report solves*

$$\frac{2A_H^2 P(Pv - \omega\kappa)}{[A_H + B_A + (Pv-\omega\kappa)^2]^2} + \frac{2Dc_\ell^2 v}{(c_\ell + 2Dv^2)^2} + c_v(v-\kappa) + c_p v = 0. \tag{21}$$

The report shares the sign of κ *and satisfies odd symmetry in type. If*

$$c_v > \frac{P\omega A_H^2}{2(A_H + B_A)^2}, \tag{22}$$

then $v^*(\kappa)$ increases strictly with κ and

$$\lim_{|\kappa|\to\infty} \frac{v^*(\kappa)}{\kappa} = \frac{c_v}{c_v + c_p} > 0. \tag{23}$$

Hence $|v^*(\kappa)| \to \infty$.

The optimized response always improves current action quality under the benchmark. Current-action loss equals

$$L_1^*(\kappa) = V + \frac{A_H M(v^*, \kappa)}{A_H + M(v^*, \kappa)}. \tag{24}$$

Relative to fact-only AI followed by unaided value implementation, personalized AI generates the strict action-quality gain

$$\boxed{V + A_H - L_1^*(\kappa) = \frac{A_H^2}{A_H + M(v^*, \kappa)} > 0.} \tag{25}$$

The uniqueness condition gives each private type one well-defined reporting choice. The proof eliminates audit exactly and shows that the minimized audit block has worst curvature $-D/2$. Condition (20) therefore guarantees a unique solution to the reduced report problem without requiring joint convexity in (v, ℓ). Economically, the result closes the first link in the paper's mechanism. Every κ maps to one report and one continuation response, which together determine the residual content carried into date 2.

Corollary 1 (User responses to provenance)**.** *Suppose $D > 0$ and (20) holds. Let $R_{vv}(v^*; \kappa, D) > 0$ denote the reduced report curvature. Then*

$$\frac{\partial v^*}{\partial D} = -\frac{2c_\ell^2 v^*(c_\ell - 2Dv^{*2})}{(c_\ell + 2Dv^{*2})^3 R_{vv}(v^*; \kappa, D)}, \tag{26}$$

and therefore

$$\operatorname{sign}\left(\frac{\partial v^*}{\partial D}\right) = -\operatorname{sign}(v^*)\operatorname{sign}(1 - 2\ell^*). \tag{27}$$

If, in addition,

$$\boxed{\underline{c} > 2D,} \tag{28}$$

then

$$\frac{\partial \ell^*}{\partial D} \geq 0, \tag{29}$$

with strict inequality for every $\kappa \neq 0$. *At* $\kappa = 0$, $v^* = \ell^* = 0$ *for every* D, *which gives a zero derivative.*

Provenance affects disclosure through the audit margin. For $s < 1$, *better persistent provenance lowers* D. *For nonzero types,*

$$\operatorname{sign}\left(\frac{\partial |v^*|}{\partial s}\right) = \operatorname{sign}(1 - 2\ell^*). \tag{30}$$

Equation (30) *has a direct behavioral interpretation. When the audit share lies below one half, stronger provenance reduces expected contamination. Optimal disclosure then rises. When the audit share exceeds one half, provenance replaces part of an already large private audit effort. Optimal disclosure can then fall. Under* (28), *stronger provenance also reduces private audit.*

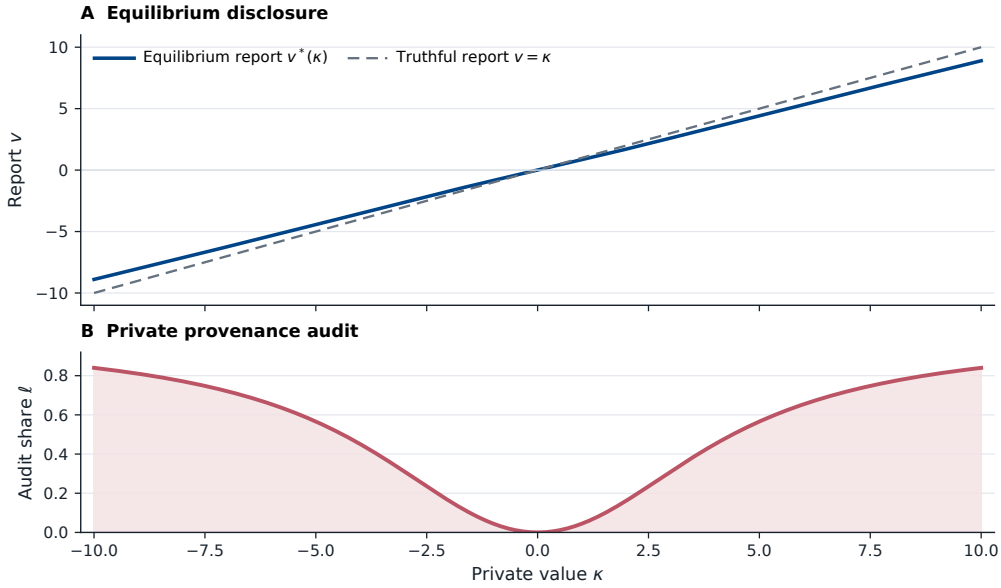


Figure 2: A single intertemporal problem determines value disclosure and provenance audit. Panel A compares the equilibrium report with truthful disclosure. Panel B shows the type-dependent audit share. The parameterization satisfies $\underline{c} > D/2$ and $\underline{c} > 2D$.

3.3 Current action gains and portable-belief harm

3.3.1 The no-AI benchmarks

The no-AI allocation provides the benchmark for both dates. Without AI, the current task uses the private factual posterior and the human value-translation technology. The corresponding loss equals

$$L_1^0 = \frac{1}{T} + A_H. \tag{31}$$

Date-2 fact-only loss equals

$$L_2^0 = \frac{1}{T}. \tag{32}$$

By Theorem 1, $L_1^*(\kappa) < L_1^0$ for every type whenever $\tau > 0$.

3.3.2 Residual harm after benchmark audit

The unclassified share of the value adjustment determines later factual distortion. Persistent provenance removes one share before the user chooses private audit. Let

$$g_0 = (1 - s)P. \tag{33}$$

Thus g_0 denotes the pre-audit unclassified value loading. After optimal audit, residual value loading equals

$$G(v) = g_0[1 - \ell^*(v)] = \frac{g_0 c_\ell}{c_\ell + 2\delta_F w^2 g_0^2 v^2}. \tag{34}$$

Date-2 factual loss equals

$$L_2^*(\kappa) = V + w^2 G(v^*)^2 v^{*2}. \tag{35}$$

For an individual type, the second term in (35) measures factual loss from treating the residual value adjustment as evidence about θ. Population loss averages this object across types: $V + w^2\mathbb{E}[G(v^*)^2 v^{*2}]$. Type-dependent reporting plus audit keep the residual loading and the report inside the expectation.

The benchmark audit technology makes residual contamination hump shaped in

report size. A normalized index makes that shape transparent. Define

$$u(v) = \frac{2\delta_F w^2 g_0^2 v^2}{c_\ell}. \tag{36}$$

The excess factual component becomes

$$H(v) = w^2 G(v)^2 v^2 = \frac{c_\ell}{2\delta_F} \frac{u(v)}{[1+u(v)]^2}. \tag{37}$$

It equals zero at $v = 0$ and converges to zero as $|v| \to \infty$. The maximum equals

$$H_{\max} = \frac{c_\ell}{8\delta_F} \tag{38}$$

when $u(v) = 1$.

Theorem 2 (Current action gains and benchmark intermediate harm)**.** *Suppose the conditions of Theorem 1 and* (22) *hold. Also assume* $g_0 > 0$ *and* $\tau > 0$*. Personalized AI strictly improves the current action for every* κ*.*

Comparison of residual contamination with the clean factual gain determines future harm. Define the clean factual improvement

$$\Delta = \frac{1}{T} - \frac{1}{T+\tau} \tag{39}$$

and

$$r = \frac{2\delta_F \Delta}{c_\ell}. \tag{40}$$

The ratio r *normalizes the clean factual gain by audit cost. A positive-measure interval of types has* $L_2^*(\kappa) > L_2^0$ *if and only if*

$$\boxed{c_\ell > 8\delta_F \Delta.} \tag{41}$$

When the condition holds, set

$$u_\pm = \frac{1 - 2r \pm \sqrt{1-4r}}{2r}, \qquad 0 < u_- < 1 < u_+, \tag{42}$$

and

$$v_\pm = \sqrt{\frac{c_\ell u_\pm}{2\delta_F w^2 g_0^2}}. \tag{43}$$

Unique cutoffs $0 < \kappa_- < \kappa_+$ *satisfy*

$$v^*(\kappa_-) = v_-, \qquad v^*(\kappa_+) = v_+.$$

Future harm occurs exactly when

$$\boxed{\kappa_- < |\kappa| < \kappa_+.} \tag{44}$$

The theorem formalizes the portfolio intuition. A related social-learning model separates state beliefs from idiosyncratic types in observed actions; small errors about the type distribution can generate large learning failures (Frick et al., 2020). Here the classification problem arises through later reuse of personalized content. A small reported liquidity need creates only a small value adjustment, so little can contaminate the later return belief. A very large reported need creates a large adjustment, but the user responds by auditing most of it under the benchmark technology. Intermediate reports combine meaningful value loading with incomplete cleaning. Those reports form the exact region where the saved recommendation can improve the original personalized decision and worsen the later fact-only decision. The benchmark audit cost $c_\ell \ell^2/2$ yields the exact result.

The factual bias first rises and then falls with type. Define the date-2 factual bias

$$b(\kappa) = wG(v^*(\kappa))v^*(\kappa). \tag{45}$$

For $\kappa > 0$,

$$b'(\kappa) = wg_0 v^{*\prime}(\kappa)\frac{c_\ell[c_\ell - 2Dv^{*2}]}{(c_\ell + 2Dv^{*2})^2}. \tag{46}$$

The bias rises below the audit midpoint and falls after audit becomes sufficiently intensive. Oddness gives the corresponding result for negative types.

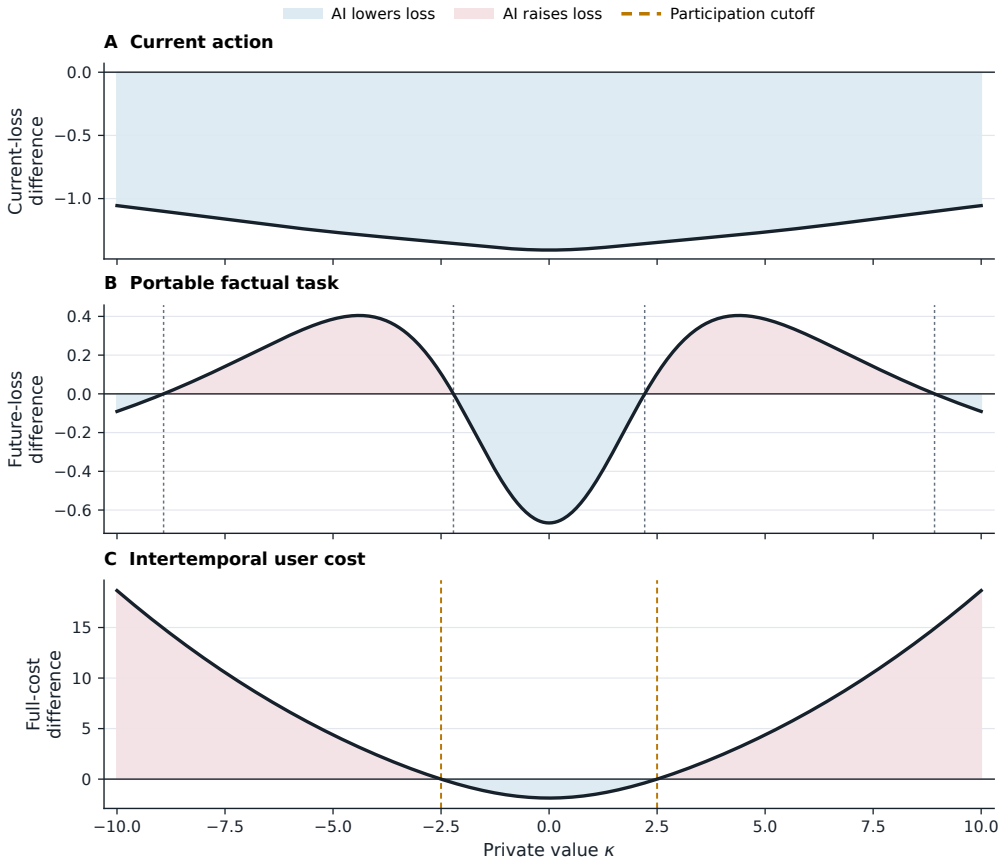


Figure 3: Each panel reports AI loss minus no-AI loss. Blue fill indicates a loss reduction; red fill indicates a loss increase. Panel B marks the exact harm-interval cutoffs. Panel C marks the zero-extra-cost participation cutoffs in gold. Full cost includes both action losses together with report-related costs and audit expenditure.

3.3.3 Full user welfare

The model treats action quality separately from full user welfare. Full welfare counts the losses from the two decisions and also charges the user for value reporting and provenance audit. Reporting costs and audit expenditure therefore enter directly. The following expressions define optimized report-related cost and optimized audit cost.

$$R_c^*(\kappa) = \frac{c_v}{2}[v^*(\kappa) - \kappa]^2 + \frac{c_p}{2}v^*(\kappa)^2, \tag{47}$$

$$A_c^*(\kappa) = \frac{c_\ell}{2}\ell^*(\kappa)^2. \tag{48}$$

Table 1: Baseline user-welfare ledger

Component	AI	No AI	AI minus no AI
Current action loss	0.65294	2.00000	−1.34706
$\delta_F\times$ future fact-task loss	0.62712	0.70000	−0.07288
Report and privacy cost	1.59105	0	1.59105
Audit cost	0.29150	0	0.29150
Full intertemporal cost	3.16261	2.70000	0.46261

Gauss-Hermite integration uses the declared theoretical type distribution. The undiscounted mean future losses equal 0.89588 with AI and 1 without AI.

Full cost conditional on AI use equals

$$\mathcal{C}^*_{AI}(\kappa) = L^*_1(\kappa) + \delta_F L^*_2(\kappa) + R^*_c(\kappa) + A^*_c(\kappa). \tag{49}$$

The no-AI outside allocation has cost

$$\mathcal{C}_0 = L^0_1 + \delta_F L^0_2 = \frac{1+\delta_F}{T} + A_H. \tag{50}$$

The exact welfare ledger therefore takes the form

$$\begin{aligned}\mathcal{C}_0 - \mathcal{C}^*_{AI}(\kappa) = [L^0_1 - L^*_1(\kappa)] + \delta_F[L^0_2 - L^*_2(\kappa)] \\ - R^*_c(\kappa) - A^*_c(\kappa).\end{aligned} \tag{51}$$

The strict date-1 action gain therefore does not determine the sign of full welfare. A user can receive a better original recommendation and still bear a higher intertemporal cost after accounting for disclosure effort plus record cleaning.

The baseline separates incidence from average loss. Table 1 reports the ledger under $\kappa \sim \mathcal{N}(0, 3^2)$. The average current action loss falls from 2 to 0.65294. The undiscounted average future fact-task loss falls from 1 to 0.89588. At the same time, 45.78% of types have higher future loss. The incidence result therefore coexists with a lower population-average future loss. Including report-related costs plus audit expenditure raises average all-user cost to 3.16261, compared with 2.70000 under no AI.

Allowing users to opt out changes which types remain exposed. An optional-use exercise lets a type observe κ and select the no-AI outside allocation at zero adoption

cost:

$$d^*(\kappa) = \mathbf{1}\{\mathcal{C}^*_{AI}(\kappa) \leq \mathcal{C}_0\}. \tag{52}$$

At the baseline, adoption has a single symmetric cutoff. The cost difference satisfies even symmetry and has one positive crossing:

$$d^*(\kappa) = 1 \quad \Longleftrightarrow \quad |\kappa| \leq 2.499931. \tag{53}$$

Selection removes much of the high-cost exposure. The implied adoption mass equals 0.595330. The population mass that adopts and has higher date-2 loss equals 0.056073. This equals 0.094188 of adopters. The all-user date-2 harmed mass equals 0.457801. The all-user mass with higher full cost equals 0.404670. Under the baseline, optional use therefore removes much of the exposure among types with high full cost. Splitting the welfare integral at the adoption cutoff gives mean optional-use cost 1.937873. The gain relative to no AI equals 0.762127. The split accounts for a kink in the lower envelope of (52).

3.3.4 Structural conditions for intermediate-type vulnerability

The intermediate-type pattern depends on the technology of source cleaning. The report rule must reach the report sizes at which contamination peaks. Audit must also become effective enough as reports grow for extreme reports to become safe again. If audit cost rises too quickly with report magnitude, residual contamination can persist in the tails. The following propositions separate these requirements.

The generalization lets audit cost vary with report size. Let generalized audit cost be

$$\frac{c_\ell}{2} h(v) \ell^2, \qquad h(v) > 0, \tag{54}$$

where $h(v)$ denotes a positive audit-cost multiplier with even symmetry and continuity. Conditional on v,

$$\ell^*_h(v) = \frac{2Dv^2}{c_\ell h(v) + 2Dv^2}, \tag{55}$$

$$H_h(v) = w^2 g_0^2 v^2 \left[\frac{c_\ell h(v)}{c_\ell h(v) + 2Dv^2} \right]^2. \tag{56}$$

Proposition 1 (Bounded-interval topology)**.** *Let $v^*_h(\kappa)$ be continuous and odd. Assume*

strict monotonicity with unbounded range. Suppose $H_h(v)$ has even symmetry and continuity. For some peak location $v_m > 0$, it rises strictly on $[0, v_m]$ and falls strictly on $[v_m, \infty)$. It satisfies $H_h(0) = 0$. Its limit as $v \to \infty$ equals zero. Whenever $\max_v H_h(v) > \Delta$, future harm occurs on a unique interval

$$\kappa_-^h < |\kappa| < \kappa_+^h.$$

Proposition 2 (Audit-cost tail classification)**.** *Suppose $D > 0$. If $h(v) \sim K|v|^m$ for $K > 0$, with m denoting the tail-growth exponent, then*

$$H_h(v) \sim \begin{cases} \dfrac{w^2 g_0^2 c_\ell^2 K^2}{4D^2} |v|^{2m-2}, & m < 2, \\ w^2 g_0^2 \left(\dfrac{c_\ell K}{c_\ell K + 2D} \right)^2 v^2, & m = 2, \\ w^2 g_0^2 v^2, & m > 2. \end{cases} \tag{57}$$

Consequently, residual contamination vanishes for $m < 1$ and diverges for $m > 1$. For $m = 1$, it converges to

$$C_1(K) = \frac{w^2 g_0^2 c_\ell^2 K^2}{4D^2} > 0. \tag{58}$$

Net future excess loss equals $H_h(v) - \Delta$. Hence large reports eventually obtain lower future loss when $m < 1$. Values $m > 1$ eventually produce higher future loss. When $m = 1$, $C_1(K) > \Delta$ gives eventual excess loss for large reports. The reverse strict inequality gives eventual future gains. If $C_1(K) = \Delta$, the stated assumptions imply only

$$H_h(v) - \Delta \longrightarrow 0.$$

They do not determine its eventual sign. For the sharper critical case, let K_* denote the critical slope and let b denote the next-order constant:

$$h(v) = K_*|v| + b + o(1), \qquad C_1(K_*) = \Delta, \tag{59}$$

then

$$H_h(v) - \Delta = \frac{2\Delta}{|v|} \left(\frac{b}{K_*} - \frac{c_\ell K_*}{2D} \right) + o(|v|^{-1}). \tag{60}$$

Thus the eventual sign at equality requires a second-order tail condition. If $|v_h^*(\kappa)| \to \infty$ as $|\kappa| \to \infty$, the same statements apply to extreme types.

The two propositions separate overall shape from tail behavior. Proposition 1 assumes a single-peaked residual-loss curve that returns to zero. Proposition 2 shows when that return occurs. When $m < 1$, the unaudited share shrinks fast enough for extreme-report contamination to vanish. When $m > 1$, the remaining share shrinks too slowly relative to report size, so harm extends to sufficiently large types. The critical case $m = 1$ depends on the limiting constant and may require the second-order condition in (60). The quadratic fidelity and privacy costs in (18) support a monotone unbounded report map under (22). Bounded disclosure can truncate the type-level harm set. Nonmonotone reporting can fragment it.

4 Platform Commitment and Provenance

4.1 Platform provision of persistent provenance

4.1.1 Ex ante commitment

The platform chooses how much causal source information will survive before any user reports a value. This commitment can change the current user response and the later reuse burden. The commitment formulation follows the information-design tradition in which policy anticipates induced behavior (Kamenica and Gentzkow, 2011; Arieli et al., 2024). Ichihashi and Smolin provide a distinct benchmark in which a designer supplies information to a seller already privately informed about buyer value (Ichihashi and Smolin, 2025). The present design variable governs persistence of source information after personalized advice has been produced. Throughout this subsection, the response technology (P, τ) remains fixed. The platform commits to s before the user reports v. Let $v_s^*(\kappa)$ and $\rho_s^*(\kappa)$ denote the optimal report and delegation. Let $\ell_s^*(\kappa)$ denote optimal audit. The user's minimized nonconstant cost has a current component and a continuation component. The split has a direct interpretation. $A(s)$ collects current implementation and reporting costs after user optimization. $B(s)$ collects future contamination together with private audit cost:

$$A(s) = \mathbb{E}\left[\frac{A_H M(v_s^*, \kappa)}{A_H + M(v_s^*, \kappa)} + \frac{c_v}{2}(v_s^* - \kappa)^2 + \frac{c_p}{2}v_s^{*2}\right], \tag{61}$$

$$B(s) = \mathbb{E}\left[\frac{c_\ell D(s) v_s^{*2}}{c_\ell + 2D(s)v_s^{*2}}\right], \qquad D(s) = \delta_F w^2 (1-s)^2 P^2. \tag{62}$$

The decomposition isolates the economic wedge. $A(s)$ collects costs that arise around the current recommendation and reporting decision. The function $B(s)$ measures continuation burden from residual portable-belief loss together with private audit cost. Let $C_s(s)$ denote the resource cost of persistent provenance. The planner minimizes

$$\mathcal{J}_1(s) = A(s) + B(s) + C_s(s). \tag{63}$$

Partial internalization scales the continuation component. A short-horizon platform values the current recommendation through $A(s)$. The parameter $\chi \in (0,1)$ measures the internalized fraction of continuation burden:

$$\mathcal{J}_\chi(s) = A(s) + \chi B(s) + C_s(s). \tag{64}$$

The platform receives payoff $-\mathcal{J}_\chi(s)$. For any internalization weight $\xi \in [\chi, 1]$, write $\mathcal{J}_\xi(s) = A(s) + \xi B(s) + C_s(s)$. The platform makes a unique ex ante design choice whenever the conditions of Theorem 3 hold.

Theorem 3 (Undercommitment to persistent provenance)**.** *Suppose A and B admit two continuous derivatives. Impose the same condition on C_s. Assume $B'(s) < 0$ on the relevant policy interval. Also assume strict convexity of $\mathcal{J}_\xi$ with an interior minimizer s^*_ξ for every $\xi \in [\chi, 1]$. Then*

$$\frac{\mathrm{d}s^*_\xi}{\mathrm{d}\xi} = -\frac{B'(s^*_\xi)}{\mathcal{J}''_\xi(s^*_\xi)} > 0. \tag{65}$$

Hence

$$\boxed{s^*_\chi < s^*_1.} \tag{66}$$

The reporting feedback can be summarized by a type-specific elasticity. For a nonzero type, define

$$u_s(\kappa) = \frac{2D(s)v^*_s(\kappa)^2}{c_\ell}$$

and let $R_{vv,s} > 0$ be reduced report curvature. The index $u_s(\kappa)$ measures normalized audit exposure at type κ. The endogenous report elasticity equals

$$(1-s)\frac{\partial \log |v^*_s(\kappa)|}{\partial s} = \frac{4D(s)[1-u_s(\kappa)]}{[1+u_s(\kappa)]^3 R_{vv,s}}. \tag{67}$$

The following inequality provides a sufficient condition for $B'(s) < 0$:

$$(1-s)\frac{\partial \log |v_s^*(\kappa)|}{\partial s} < 1 \tag{68}$$

for almost every nonzero type. A primitive uniform sufficient condition takes the form

$$\boxed{\underline{c} > \frac{9}{2}D(s)} \tag{69}$$

throughout the relevant policy interval.

If Theorem 1 holds throughout the policy interval, the theorem delivers a unique backward-induction outcome for each $\xi \in [\chi, 1]$.

The ordering follows from who bears the later cost. A planner assigns full weight to $B(s)$, while the platform assigns weight χ. When $B'(s) < 0$, an increase in provenance reduces the continuation burden, so a designer with a larger internalization weight chooses more provenance. Disclosure can respond to the policy at the same time. Condition (68) ensures that this behavioral response remains smaller than the direct reduction in unclassified loading. The primitive curvature condition provides one transparent sufficient condition for that property.

4.1.2 A tractable fixed-report benchmark

The fixed-report benchmark strips away the disclosure feedback so the direct provenance incentive can be seen on its own. Hold the report distribution fixed and let $g \in [0, P]$ be the pre-audit value loading left unclassified. Persistent provenance satisfies $s = 1 - g/P$. For a realized report v, minimized continuation burden equals

$$\Psi(g, v) = \frac{c_\ell \delta_F w^2 g^2 v^2}{c_\ell + 2\delta_F w^2 g^2 v^2}. \tag{70}$$

Expected continuation burden equals $\overline{\Psi}(g) = \mathbb{E}[\Psi(g, v)]$. The parameter $\xi \in (0, 1]$ measures the share internalized by the designer. The coefficient $c_s > 0$ governs quadratic provenance cost. The designer minimizes

$$\widetilde{\mathcal{J}_\xi}(g) = \xi \overline{\Psi}(g) + \frac{c_s}{2}(P - g)^2. \tag{71}$$

Proposition 3 (Direct provenance margin)**.** *Suppose*

$$P > 0, \qquad \xi \in (0, 1], \qquad 0 < \mathbb{E}[v^2] < \infty, \tag{72}$$

and

$$c_s > \frac{\delta_F w^2 \mathbb{E}[v^2]}{2}. \tag{73}$$

Then (71) *has strict convexity and admits a unique interior solution* $g^*_\xi \in (0, P)$*. Moreover,*

$$\frac{\mathrm{d}g^*_\xi}{\mathrm{d}\xi} < 0. \tag{74}$$

For $0 < \chi < 1$*,*

$$\boxed{g^*_\chi > g^*_1, \qquad s^*_\chi < s^*_1.} \tag{75}$$

The convexity bound also guarantees an interior solution. The positive second moment and $\xi > 0$ make the derivative at $g = P$ positive; together with (73), these conditions yield the stated interior solution.

Corollary 2 (Binary automatic metadata)**.** *Suppose the designer chooses* $m \in \{0, 1\}$ *before reporting, where* $m = 1$ *activates an automatic provenance system with resource cost* $c_m \geq 0$*. Let* A_m *and* B_m *denote the resulting current and continuation costs after user optimization. The planner chooses automatic metadata if and only if*

$$(A_0 + B_0) - (A_1 + B_1) \geq c_m. \tag{76}$$

A platform with continuation internalization $\xi \in [0, 1]$ *chooses it if and only if*

$$(A_0 + \xi B_0) - (A_1 + \xi B_1) \geq c_m. \tag{77}$$

Continuation burden provides a partial welfare statistic. Metadata may also change prompt supply and current implementation. Audit provides another channel.

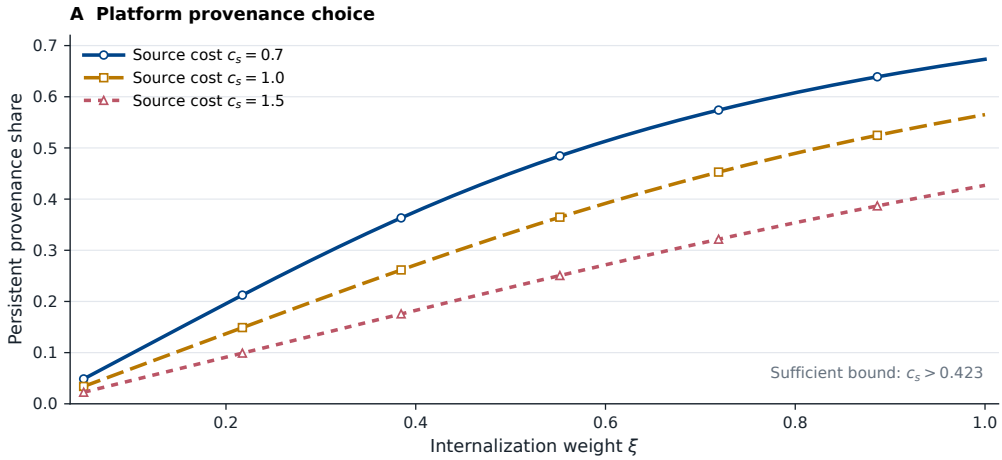


Figure 4: Fixed-report provenance benchmark across internalization weights for three source-cost values. The benchmark holds the report distribution fixed at $v \sim \mathcal{N}(0, 1.65^2)$. The vertical axis reports $s^*_\xi = 1 - g^*_\xi/P$. This figure solves the fixed-report benchmark. Every displayed source cost exceeds the sufficient convexity bound.

5 Awareness Boundaries and Quantitative Illustrations

5.1 Causal-awareness boundary

The causal-awareness boundary states exactly when the mechanism disappears. Research on misspecified models emphasizes that inference depends on the represented information environment (Bohren and Hauser, 2025). Spiegler provides a related source-classification example in which users fail to distinguish raw data from imputed data (Spiegler, 2017). In the portfolio example, a saved note that still identifies the liquidity-based tilt lets the later user separate that tilt from evidence about returns before forming the fact-only belief. If a complete causal record remains available at date 2, the user can condition on v and remove the value adjustment. A causally aware user can also fall back on the decision rule available without the AI signal. Additional factual information therefore weakly lowers minimum Bayes risk, consistent with the Blackwell comparison under correct representation (Blackwell, 1953).

Lemma 2 (Causal-awareness boundary)**.** *Let $\mathcal{E}_0$ be a baseline experiment about θ. Let $\mathcal{E}_1$ Blackwell dominate $\mathcal{E}_0$ conditional on retained value data. Suppose the date-2*

decision maker observes the retained value report and its loading. The decision maker knows the true joint response law. The action set satisfies measurability and contains every decision rule available under $\mathcal{E}_0$. For any loss with finite Bayes risk, minimum Bayes risk under $\mathcal{E}_1$ does not exceed the corresponding risk under $\mathcal{E}_0$. Correct causal awareness of the value origin generates no preference-belief laundering.

The boundary turns on what the later decision maker can still represent. Assumption 1 therefore does substantive work. It specifies a compressed record and causal unawareness. Retention of the historical type plus the strictly monotone report map permits recovery of the realized report. The benchmark contamination term can then be removed. Awareness of the source category can also support statistical correction when the realized report remains unavailable.

5.2 Capability on feasible technology paths

Greater factual capability has two channels in the portable record. It improves the clean estimate of θ while potentially increasing the posterior weight placed on residual value content that the later model misclassifies as factual. Related GEB work also obtains nonmonotone performance in human-AI integration through strategic reliance (Lagziel and Tsodikovich, 2025). The channel here comes from provenance loss. The main theorem compares the two dates at a fixed feasible design. This subsection varies factual precision after disclosure and audit adjust endogenously.

The comparative static varies factual precision while holding personalization and provenance fixed. Fix (P, s) and let factual precision τ change. Let $\mathcal{C}^*(\kappa; \tau)$ be the minimized two-date objective in (12). The envelope theorem gives a structural derivative.

Proposition 4 (Capability after user optimization)**.** *At any optimal response satisfying* (20)*,*

$$\boxed{\frac{\mathrm{d}\mathcal{C}^*}{\mathrm{d}\tau} = -\frac{1+\delta_F}{(T+\tau)^2} + \frac{2T\delta_F g_0^2 \tau}{(T+\tau)^3} v^{*2}(1-\ell^*)^2.} \tag{78}$$

The first term captures the clean-information benefit across the two tasks. The second term captures increased factual reliance on residual value content. Capability lowers total loss if and only if the first term dominates.

Equation (78) makes the tradeoff explicit after user optimization. The first term

captures the clean-information gain, while the second captures additional reliance on residual value content. The sign depends on how much contamination survives when factual precision rises. Equation (78) holds for fixed (P, s). Figure 5 uses $\alpha = \beta = 1/\sqrt{2}$ and $\tau_I = 4$. Along this feasible primitive path, response-noise variance falls while $P = 1$ and $\tau = [\tau_I^{-1} + 2\sigma_\zeta^2]^{-1}$. A capability change that shifts the personalization frontier also has a direct P effect.

Figure 5 isolates future factual loss along the same feasible path. The plotted curves show how the clean-information channel and the residual-reliance channel combine for different private-value magnitudes, while (78) gives the corresponding derivative of optimized total loss.

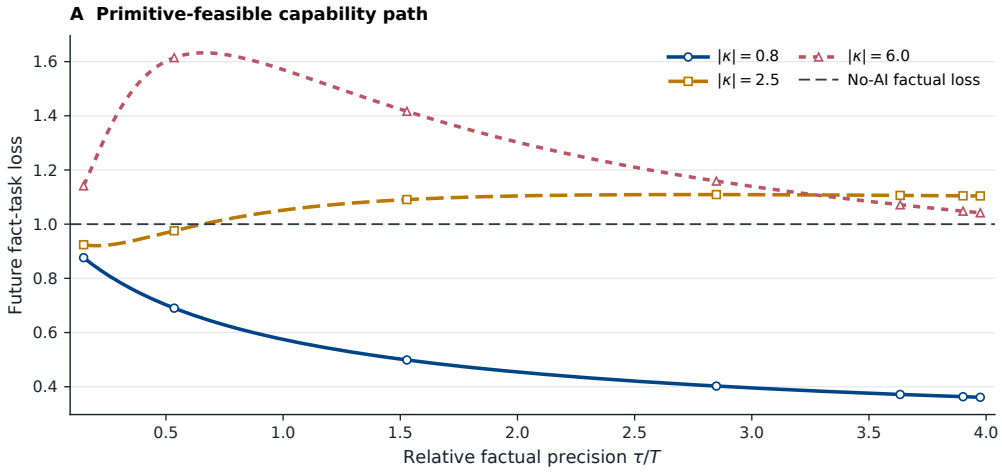


Figure 5: Future factual loss along an explicit capacity-feasible noise-improvement path. Factual precision changes because primitive response noise falls while $\alpha = \beta = 1/\sqrt{2}$ keeps $P = 1$. Curves correspond to $|\kappa| \in \{0.8, 2.5, 6\}$. The horizontal line marks the no-AI factual loss.

5.3 Imperfect labels

Provenance improves inference only through accurate source classification. Sparse internal models can omit distinctions that matter for choice (Gabaix, 2014). Procedural inference can likewise merge raw and imputed inputs into one practical belief (Spiegler, 2017). A coarse label attached to an entire recommendation can cause a later user to discount valid return evidence together with the liquidity adjustment. The relevant design problem therefore preserves the causal role of personalized content while retaining genuine factual evidence. The following two-signal calculation defines every

precision explicitly. It also includes the prior. Let the independent factual signals be

$$x_H = \theta + \epsilon_H, \qquad x_A = \theta + \epsilon_A,$$

with precisions τ_H and $\tau_A = \tau$. Labeling changes the effective weights placed on these factual signals. Let $u(s) \geq 0$ and $\mu(s) \geq 0$ be the effective precision multipliers retained for the human and AI factual signals after labeling. The represented estimator takes the form

$$\widehat{\theta}_s = \frac{u(s)\tau_H x_H + \mu(s)\tau_A x_A}{\tau_0 + u(s)\tau_H + \mu(s)\tau_A}.$$

The resulting MSE measures the factual cost of imperfect weighting. Its true Bayes MSE equals

$$L_{\text{label}}(s) = \frac{\tau_0 + u(s)^2\tau_H + \mu(s)^2\tau_A}{[\tau_0 + u(s)\tau_H + \mu(s)\tau_A]^2}. \tag{79}$$

The regret identity compares imperfect weighting with correct source use. Relative to correct source use,

$$\begin{aligned} L_{\text{label}}(s) - \frac{1}{\tau_0 + \tau_H + \tau_A} = {} & \frac{\tau_0\tau_H[u(s)-1]^2 + \tau_0\tau_A[\mu(s)-1]^2}{(\tau_0 + \tau_H + \tau_A)[\tau_0 + u(s)\tau_H + \mu(s)\tau_A]^2} \\ & + \frac{\tau_H\tau_A[u(s)-\mu(s)]^2}{(\tau_0 + \tau_H + \tau_A)[\tau_0 + u(s)\tau_H + \mu(s)\tau_A]^2}. \end{aligned} \tag{80}$$

Accurate provenance preserves both factual weights. It sets both multipliers to one. Indiscriminate suppression creates regret through underweighting and unequal weighting. The design target requires accurate causal classification.

5.4 Baseline parameterization

The numerical section illustrates the mechanisms under one declared parameterization and then varies the key primitives. The sensitivity exercises hold the baseline fixed and carry no empirical calibration interpretation. The numerical exercises use

$$(A_H, B_A, P, \omega, c_v, c_p, c_\ell, \delta_F, T, \tau, s, \sigma_\kappa) = (1, 0.35, 1, 1, 3, 0.4, 6, 0.7, 1, 2, 0.2, 3).$$

The final entry σ_κ denotes the standard deviation of the private-value distribution. Three steps map the primitives into population harm. The ratio $c_\ell/(8\delta_F\Delta)$ determines whether a harmful *report* interval exists. The optimal report map then translates that

interval into a *type* interval. The type distribution determines how much probability lies inside the type interval. The analysis reports all-user harm separately from average full cost. Optional-use exposure measures harmed adopters.

5.5 Comparative statics

The comparative statics ask which parts of the mechanism move the harmed set and which parts only move its population mass. The exercises vary audit incentives and report following, then change provenance and factual reliance. Structural exercises separately alter audit-cost tail growth and the type distribution. The following ratios summarize the baseline parameterization:

$$\begin{aligned} \lambda_\ell &= \frac{c_\ell}{8\delta_F \Delta} = \frac{6}{8(0.7)(2/3)} \simeq 1.607 > 1, & \lambda_v &= \frac{c_v}{c_v + c_p} = \frac{3}{3 + 0.4} \simeq 0.882, \\ s &= 0.2, & q_\tau &= \frac{\tau}{T} = 2. \end{aligned} \tag{81}$$

The ratio λ_ℓ measures audit slack. The ratio λ_v measures report following. The ratio q_τ measures relative factual precision. The first ratio places the model strictly inside the region where a harmful report interval exists. The second makes the tail of the report rule track type closely. Low persistent provenance leaves 80% of the intended value loading initially unclassified. The ratio $\tau/T = 2$ implies posterior reliance $w = 2/3$. These numbers place the baseline inside the active mechanism region. High audit cost relative to the clean factual gain permits harm. Reports still track private values fairly closely. Under a centered normal type distribution with standard deviation 3, the resulting all-user harmed mass equals 0.458.

Each exercise changes one mechanism at a time. Table 2 reports the four comparative statics. The report-following exercise holds $c_v + c_p = 3.4$ fixed, which allows the ratio to change without altering total quadratic report-cost curvature. The relative-precision exercise fixes $T = 1$ and varies τ. Every parameter value triggers reoptimization of reporting plus audit.

Higher factual precision raises the clean gain Δ. It also raises w, the weight attached to residual value content. The harmed mass consequently varies nonmonotonically over the values shown. Factual capability has no unconditional harm ranking in this model. The provenance exercise has a more direct policy reading. Raising s from 0.2 to 0.8 reduces the harmed mass from 0.458 to 0.005. Complete provenance removes

Table 2: Comparative statics for four parameter groups

Parameter group	Values	All-user harm / adopter harm / mean full-cost excess	Economic role
Audit slack λ_ℓ	$0.8, 1, 1.607, 2.4$	Fixed: 0, 0, 0.458, 0.495 Adopt: 0, 0, 0.056, 0.083 Full: 0.264, 0.326, 0.463, 0.578	Harmful report interval exists iff the ratio exceeds one.
Report following λ_v	$0.5, 0.7,$ $0.882, 0.95$	Fixed: 0.181, 0.359, 0.458, 0.481 Adopt: 0, 0, 0.056, 0.185 Full: 2.584, 2.060, 0.463, -0.386	Changes the report-to-type map; leaves the existence condition unchanged.
Persistent provenance s	$0, 0.2, 0.5, 0.8, 1$	Fixed: 0.525, 0.458, 0.253, 0.005, 0 Adopt: 0.102, 0.056, 0, 0, 0 Full: 0.647, 0.463, 0.145, -0.156, -0.240	Moves type cutoffs for $s < 1$; complete labeling removes contamination.
Relative precision q_τ	$0.5, 1, 2, 4$	Fixed: 0.376, 0.436, 0.458, 0.427 Adopt: 0, 0.029, 0.056, 0.060 Full: 0.603, 0.538, 0.463, 0.386	Raises both clean gain and reliance on residual value content.

Entries in each outcome cell follow the value order in the second column. "All-user harm" reports the date-2 harmed mass across the reference population. "Adopter harm" uses the zero-extra-cost outside option. Mean full-cost excess reports AI minus no AI for the all-user calculation. Masses use the centered normal type distribution with $\sigma_\kappa = 3$.

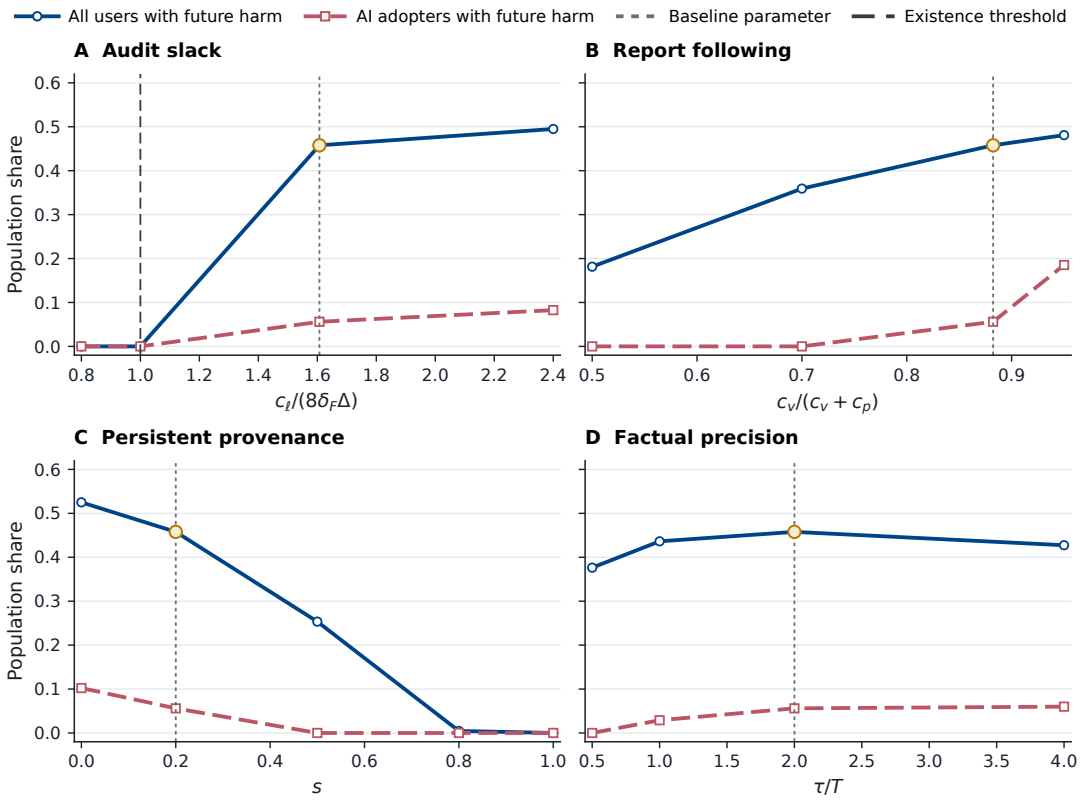


Figure 6: Comparative statics for four parameter groups. Solid blue lines show the all-user date-2 harmed mass. Dashed red lines show the mass of AI adopters with higher date-2 loss. Dotted vertical lines mark baseline parameters. The black dashed line in Panel A marks the harm-existence threshold.

Table 3: Type-scale dependence of the model-implied harmed mass

σ_κ	0.5	0.75	1	1.5	2	3	5	10	20
Harmed mass	0.00001	0.00317	0.02691	0.14015	0.26853	0.45780	0.58369	0.45254	0.25634

The calculation holds the endogenous cutoffs and all model primitives fixed. Only the centered normal type distribution's standard deviation changes. The entries report model-implied probability masses.

residual contamination in the baseline calculation.

Audit-cost tail growth determines whether residual contamination disappears for extreme reports. The exponent m in $h(v) = (1 + v^2)^{m/2}$ changes the tail shape of residual contamination. The residual vanishes for $m < 1$. It converges to a positive constant for $m = 1$. It diverges for $m > 1$. For $m = 1$, the constant must be compared with Δ. Equality also requires the second-order condition in (60).

Type dispersion determines how much population mass lies inside the endogenous harm interval. The type distribution leaves the benchmark report-space theorem unchanged. It determines the population mass between the endogenous type cutoffs. Table 3 shows a low normal harmed mass at small type dispersion. The largest displayed mass occurs at an intermediate scale. It then falls as probability moves beyond the upper cutoff. The standard deviation 3 baseline produces relatively high harmed mass within the displayed scale path. At this standard deviation, the implied masses equal 0.337 for Laplace and 0.407 for logistic. The Student-t_5 mass equals 0.373.

The appendix broadens the numerical support. Appendix E.3 varies the remaining user-side primitives one at a time. It also reports a 135-cell exercise. The grid spans total report cost and δ_F. It varies absolute precision across five provenance levels. Provenance-stratified summaries report its support. Figure 4 varies the platform design parameters. Figure 5 follows a primitive-feasible technology path.

6 Scope and Limitations

The main results rely on a specific representation of what survives across decisions. This section states the principal scope restrictions and indicates which conclusions depend on them.

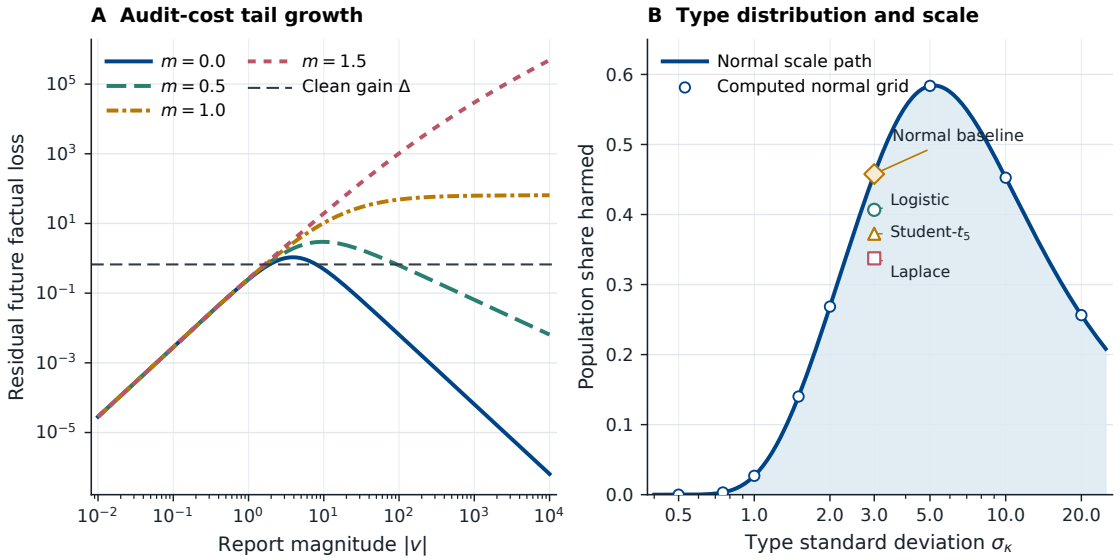


Figure 7: Two structural dimensions. Panel A changes audit-cost tail growth and plots residual future factual loss against the clean gain. For $m = 1$, the limiting constant must be compared with Δ; equality requires the second-order tail condition. Panel B keeps the benchmark type cutoffs fixed. It traces the normal scale path and marks alternative centered distributions with standard deviation 3.

6.1 Belief representation

The belief-representation assumption provides the central behavioral boundary. Bounded-memory models show how compressed information can alter later processing (Wilson, 2014; Fudenberg et al., 2024). Misspecified-learning research studies inference under an incorrect model of the information environment (Bohren and Hauser, 2025), while social-learning results show the sensitivity of state inference to misperceived type structure (Frick et al., 2020). The core mechanism applies to reuse settings where the saved record keeps part of the recommendation while losing the reason behind that content. Assumption 1 represents this loss through the likelihood $r_2 = \theta + \varepsilon_q$. The represented model also omits inference from the induced report distribution. A later decision maker who retains the value-origin category can use that distribution for statistical correction. The laundering results apply to the behavioral representation stated in the assumption.

6.2 Correlated state and value

Under joint Gaussianity with correlation between (θ, κ), the rational benchmark conditions on both variables. Write

$$\theta = \mathbb{E}[\theta \mid \kappa] + \theta^{\perp}, \qquad T_{\theta|\kappa} = \mathrm{Var}(\theta \mid \kappa)^{-1} + \tau_H.$$

For a jointly Gaussian pair, the conditional variance does not vary with the realized value of κ. The same holds for $T_{\theta|\kappa}$. When κ remains available at reuse, the baseline calculation applies to $\theta^{\perp}$ after replacing T with conditional posterior precision. Loss of κ requires a date-2 posterior based on the joint law of (x, r_2). Reporting then incorporates the factual information contained in κ. A non-Gaussian joint law can generate a nonlinear conditional mean that exceeds the scalar-precision representation. The laundering object isolates value-derived content beyond the rational informational channel.

6.3 Implementation timing and covariance

The benchmark selects ρ before realization of η_H plus η_A. Selection after both realizations produces an ex post mixing problem with a realization-dependent weight. Correlated implementation errors with covariance C satisfy

$$C^2 \leq A_H B_A.$$

For a nondegenerate covariance matrix and mismatch, the interior pre-realization rule becomes

$$\rho^* = \frac{A_H - C}{A_H + M - 2C},$$

projected onto $[0, 1]$. Covariance feasibility guarantees a nonnegative denominator. At $A_H = B_A = C$ and $Pv - \omega\kappa = 0$, every delegation weight produces the same implementation loss. The benchmark formula uses the zero-covariance case stated in Section 2.

6.4 Platform objective and participation

The platform theorem provides an intensive-margin result for a fixed participating population, where the exposure share χ measures the reduced-form weight placed on continuation burden. A richer profit model with demand or liability could generate this parameter.

Endogenous participation adds a composition channel. Let $d_s(\kappa) \in [0,1]$ denote the adoption probability of type κ. Let $b_s(\kappa)$ denote that adopter's continuation burden. Endogenous participation adds the composition term

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}[d_s(\kappa)b_s(\kappa)] = \mathbb{E}[d_s(\kappa)\partial_s b_s(\kappa)] + \mathbb{E}[(\partial_s d_s(\kappa))b_s(\kappa)]. \tag{82}$$

A sufficiently small composition effect preserves the platform ordering. A negative full derivative also preserves the ordering. A sufficient bound takes the form

$$\mathbb{E}[(\partial_s d_s(\kappa))b_s(\kappa)] < -\mathbb{E}[d_s(\kappa)\partial_s b_s(\kappa)].$$

The optional-use calculation in (52) supplies a user-side outside-option exercise at the baseline design. A platform demand model would determine $d_s(\kappa)$ across provenance policies.

6.5 Technology and numerical exercises

The remaining limitations concern technology and calibration. The response technology remains fixed apart from the feasible capability paths stated in the text. Adaptive dialogue or platform training can add dynamic feedback. Relational support can change factual reliance. The numerical exercises evaluate theoretical type distributions under declared parameters. Their probability masses represent model implications without population calibration.

7 Conclusion

Personalized advice can remain economically useful at the original decision and become misleading after loss of its rationale at reuse. The mechanism begins with a legitimate private value that changes the appropriate date-1 action. The AI translates the reported

value into a recommendation alongside factual information. If the recommendation later survives without enough source information, residual value-derived content enters a fact-only posterior. The paper calls this source-classification error preference-belief laundering. A complete causal record preserves the value origin and removes the distortion.

The user results derive the distortion from endogenous disclosure plus audit. The report determines how much private value enters the recommendation. Delegation determines implementation of the current value adjustment. Audit determines the amount of unlabeled value content cleaned before the record becomes portable. Under the benchmark audit technology, small reports create little residual loading and large reports induce intensive cleaning. Intermediate reports can therefore carry the largest factual contamination. Sufficiently costly audit produces an exact interval of types with improved current action and worse later fact-only performance. Disclosure cost plus audit expenditure can move full welfare differently from action quality.

Persistent provenance changes both the later record and the user's earlier behavior. By preserving why personalized content entered the recommendation, stronger provenance reduces the amount that can be misclassified at reuse and changes the return to private audit. The user can respond by changing value disclosure. A planner internalizes the full continuation burden created by these choices, while a platform with exposure share $\chi < 1$ internalizes a smaller share. Under the theorem's regularity conditions, that wedge leads the platform to choose less persistent provenance.

The design implication favors preservation of causal source information whenever personalized recommendations can travel across decisions. Useful provenance identifies which content reflects the user's value and keeps that rationale attached to the saved record. It also preserves genuine factual evidence rather than suppressing the recommendation as a whole. In this setting, evaluating personalized AI requires attention to both the quality of the original action and the quality of the factual belief that later survives with the recommendation.

A Proofs for the User Response

A.1 Proof of the feasibility frontier

The feasibility frontier follows by eliminating the primitive signal weight. From (4),

$$\beta(\tau) = \sigma_\zeta \sqrt{\frac{\tau \tau_I}{\tau_I - \tau}}.$$

The capacity constraint gives $P^2 = \alpha^2/\beta^2 \leq (1-\beta^2)/\beta^2$. Substitution yields (5). Setting $P = \omega$ and solving the capacity condition for τ gives (6). □

A.2 Proof of the unified user response

For a fixed report, the user problem separates into delegation and audit. Conditional on v, value-translation loss equals

$$\rho^2 M(v,\kappa) + (1-\rho)^2 A_H.$$

Strict convexity makes the first-order condition sufficient and yields (16). Substitution yields

$$f(Pv - \omega\kappa) = \frac{A_H M(v,\kappa)}{A_H + M(v,\kappa)}.$$

Conditional on v, the audit problem takes the form

$$Dv^2(1-\ell)^2 + \frac{c_\ell}{2}\ell^2.$$

Equation (17) gives its solution. Substitution gives the audit term in (18).

Global uniqueness follows from a lower bound on reduced curvature. Let $t = Pv - \omega\kappa$. The implementation term satisfies

$$f''(t) = \frac{2A_H^2[A_H + B_A - 3t^2]}{[A_H + B_A + t^2]^3},$$

with global minimum

$$\min_t f''(t) = -\frac{A_H^2}{2(A_H + B_A)^2}.$$

Let $q(v) = c_\ell D v^2/(c_\ell + 2Dv^2)$ denote the minimized audit term. Its second derivative

equals

$$q''(v) = \frac{2Dc_\ell^2(c_\ell - 6Dv^2)}{(c_\ell + 2Dv^2)^3}.$$

Writing $u = 2Dv^2/c_\ell$ gives

$$q''(v) = 2D\frac{1-3u}{(1+u)^3},$$

whose global minimum equals $-D/2$ at $u = 1$. Therefore

$$R_{vv}(v;\kappa,D) \geq \underline{c} - \frac{D}{2} > 0. \tag{83}$$

Strict curvature and coercivity give a unique report. Coercivity follows from $c_v + c_p > 0$, which gives the reduced objective a unique minimizer under strict curvature. Differentiation gives (21). The conditional best responses ρ^* and ℓ^* then yield the unique full response.

The first-order condition also pins down report sign and monotonicity. At $v = 0$, the report FOC takes a negative value for $\kappa > 0$ and a positive value for $\kappa < 0$. Strict increase of the FOC in v gives the unique root the same sign as type. Symmetry gives oddness. Implicit differentiation gives

$$\frac{\mathrm{d}v^*}{\mathrm{d}\kappa} = \frac{c_v + P\omega f''(Pv^* - \omega\kappa)}{R_{vv}(v^*;\kappa,D)}.$$

Condition (22) makes the numerator positive.

The report rule approaches linear growth in type. The report FOC can be written as

$$(c_v + c_p)v^* - c_v\kappa = -Pf'(Pv^* - \omega\kappa) - q'(v^*).$$

Both terms on the right remain globally bounded. Hence

$$v^*(\kappa) = \frac{c_v}{c_v + c_p}\kappa + O(1),$$

which proves (23). Finally,

$$A_H - \frac{A_H M}{A_H + M} = \frac{A_H^2}{A_H + M} > 0,$$

which proves (25). □

A.3 Proof of the provenance-disclosure comparative statics

The report response to contamination follows from implicit differentiation. Let $F(v, \kappa, D) = R_v(v; \kappa, D)$. Direct differentiation gives

$$F_D = \frac{2c_\ell^2 v(c_\ell - 2Dv^2)}{(c_\ell + 2Dv^2)^3}.$$

Since $F_v = R_{vv} > 0$, the implicit-function theorem gives (26) and (27).

The total audit response requires the unreduced problem. Return to the unreduced (v, ℓ) objective after substituting delegation. At the optimal audit choice, define

$$\begin{aligned} H_{vv} &= P^2 f''(Pv - \omega\kappa) + c_v + c_p + 2D(1-\ell)^2, \\ H_{\ell\ell} &= c_\ell + 2Dv^2, \\ H_{v\ell} &= -4Dv(1-\ell). \end{aligned}$$

The Hessian links the unreduced problem to reduced curvature. The determinant satisfies $\det H = H_{\ell\ell} R_{vv} > 0$. Differentiating the two first-order conditions gives

$$\frac{\partial \ell^*}{\partial D} = \frac{2v^{*2}(1-\ell^*)[H_{vv} - 4D(1-\ell^*)^2]}{\det H}. \tag{84}$$

The stronger curvature condition makes the audit response monotone. Moreover,

$$H_{vv} - 4D(1-\ell)^2 \geq \underline{c} - 2D(1-\ell)^2.$$

Condition (28) makes this expression positive. The remaining factors in (84) remain positive for $v^* \neq 0$, while the derivative equals zero at $v^* = 0$. Since $D_s = -2D/(1-s)$, (30) and the audit response to s follow. □

B Proofs for Portable-Belief Harm

B.1 Proof of the benchmark coexistence theorem

The hump shape follows directly after substituting optimal audit. After substituting (17),

$$1 - \ell^*(v) = \frac{c_\ell}{c_\ell + 2Dv^2}, \qquad D = \delta_F w^2 g_0^2.$$

This gives (37). The function $u/(1+u)^2$ reaches its maximum $1/4$ at $u=1$, which proves (38).

The harm interval comes from comparing the hump with the clean factual gain. The future loss exceeds no AI exactly when

$$\frac{c_\ell}{2\delta_F}\frac{u}{(1+u)^2} > \Delta.$$

With $r = 2\delta_F\Delta/c_\ell$, the inequality becomes $u/(1+u)^2 > r$. A nonempty solution exists exactly when $r < 1/4$, yielding (41). The equality condition takes the form

$$ru^2 + (2r-1)u + r = 0,$$

whose roots follow (42). Translating the roots into report magnitudes gives (43). Strict monotonicity and unboundedness of v^* give unique type cutoffs. The current gain follows from (25) and $V < 1/T$. □

Direct differentiation of residual loading gives the bias slope. Differentiating

$$b(\kappa) = wg_0\frac{c_\ell v^*(\kappa)}{c_\ell + 2Dv^*(\kappa)^2}$$

gives (46).

B.2 Proofs of the bounded-interval and audit-tail propositions

Generalized audit retains the same one-dimensional minimization structure. Minimizing

$$Dv^2(1-\ell)^2 + \frac{c_\ell}{2}h(v)\ell^2$$

with respect to ℓ gives (55). Substitution into the undiscounted date-2 contamination loss gives (56).

The bounded interval follows from two crossings of the residual-loss curve. Under the conditions in Proposition 1, every level strictly below $\max_v H_h(v)$ and above zero has exactly two positive report-space crossings. An unbounded report map with continuity plus strict monotonicity transfers those crossings into unique type cutoffs. This proves the bounded-interval statement.

The dominant denominator term determines tail behavior. If $h(v) \sim K|v|^m$ and

$m < 2$, the term $2Dv^2$ dominates the denominator, which gives the first line of (57). At $m = 2$, both denominator terms have order v^2. For $m > 2$, the audit-cost term dominates and the residual fraction converges to one. Setting $m = 1$ in the first line gives (58). Future excess loss relative to no AI equals $H_h(v) - \Delta$, so strict comparison of $C_1(K)$ with Δ determines the eventual sign. Equality determines only the zero limit.

The equality case requires one more term in the tail expansion. For the sharper critical statement, take $v > 0$ without loss of generality and write $h(v) = K_* v + b + o(1)$. Expanding the residual factual loading gives

$$\frac{w g_0 c_\ell v h(v)}{c_\ell h(v) + 2Dv^2} = \frac{w g_0 c_\ell K_*}{2D} \left[1 + \frac{1}{v} \left(\frac{b}{K_*} - \frac{c_\ell K_*}{2D} \right) + o(v^{-1}) \right].$$

Squaring and using $C_1(K_*) = \Delta$ yields (60). A zero displayed coefficient requires still higher-order terms. Divergence for $m > 1$ and unboundedness of the report map imply eventual extreme-type harm. □

C Proof of the Capability Proposition

The capability derivative separates clean information from residual contamination. For fixed (P, s), the minimized two-date objective includes clean variance $(1 + \delta_F)/(T + \tau)$ and contamination coefficient

$$D(\tau) = \delta_F g_0^2 \left(\frac{\tau}{T + \tau} \right)^2.$$

The envelope theorem removes derivatives of the optimized report and audit. Differentiation gives

$$D'(\tau) = \frac{2T \delta_F g_0^2 \tau}{(T + \tau)^3},$$

and substitution yields (78). □

D Proofs for Platform Provenance Design

D.1 Proof of the integrated platform theorem

The platform wedge follows from the design first-order condition. The first-order condition takes the form

$$A'(s_\xi^*) + \xi B'(s_\xi^*) + C_s'(s_\xi^*) = 0.$$

Implicit differentiation gives (65). Strict convexity makes the denominator positive, while $B' < 0$ makes the numerator positive.

The sufficient condition controls the reporting feedback. For the endogenous-report condition, write $u = 2Dv^2/c_\ell$. Equation (26) becomes

$$\frac{\partial v^*}{\partial D} = -\frac{2v^*(1-u)}{(1+u)^3 R_{vv}}.$$

Using $D_s = -2D/(1-s)$ gives (67). Conditional continuation burden equals $c_\ell x/(c_\ell + 2x)$ with $x = Dv^2$, which increases strictly in x. Condition (68) implies $x_s < 0$ almost everywhere and therefore $B'(s) < 0$.

The primitive curvature bound makes the elasticity small enough. If $u \geq 1$, the elasticity in (67) does not exceed zero. If $u < 1$, then

$$\frac{4D(1-u)}{(1+u)^3 R_{vv}} \leq \frac{4D}{\underline{c} - D/2} < 1$$

under (69). □

D.2 Proof of the fixed-report benchmark

The fixed-report proof starts from the curvature of continuation burden. For fixed v, let $a = \delta_F w^2 v^2$. Then

$$\Psi(g, v) = \frac{c_\ell a g^2}{c_\ell + 2ag^2}.$$

Differentiation gives

$$\Psi_g = \frac{2ac_\ell^2 g}{(c_\ell + 2ag^2)^2},$$
$$\Psi_{gg} = -\frac{2ac_\ell^2(6ag^2 - c_\ell)}{(c_\ell + 2ag^2)^3}.$$

The curvature bound and endpoint derivatives deliver an interior optimum. Writing $t = 2ag^2/c_\ell$ gives $\Psi_{gg} \geq -a/2$. Hence

$$\overline{\Psi}''(g) \geq -\frac{\delta_F w^2 \mathbb{E}[v^2]}{2}.$$

Condition (73) implies strict convexity. At $g = 0$,

$$\widetilde{\mathcal{J}}'_\xi(0) = -c_s P < 0.$$

At $g = P$,

$$\widetilde{\mathcal{J}}'_\xi(P) = \xi \overline{\Psi}'(P) > 0$$

because $\xi > 0$ and $\Pr(v \neq 0) > 0$. Thus the unique minimizer lies in the interior. Implicit differentiation of

$$\xi \overline{\Psi}'(g^*_\xi) + c_s(g^*_\xi - P) = 0$$

gives $\mathrm{d}g^*_\xi/\mathrm{d}\xi < 0$. □

D.3 Proof of the binary metadata corollary

The binary choice follows from comparing the optimized objective values. The planner compares objective values $A_0 + B_0$ and $A_1 + B_1 + c_m$. Comparing them yields (76). Replacing B_m by ξB_m yields (77). Because A_m and B_m use post-optimization values, both comparisons include every induced response. □

E Correct-Model and Labeling Details

E.1 Proof of the correct-model boundary

Correct causal representation restores the usual Blackwell comparison. A Blackwell-more-informative experiment can be garbled to reproduce the less informative experiment. A user with the correct causal model and retained value data can implement every decision rule available before the improvement, including a rule that ignores the AI response. The minimum over a weakly larger rule set cannot have higher Bayes risk. Correctly represented coefficient uncertainty enters the likelihood and cannot create an unmodeled systematic value loading. □

E.2 Proof of the imperfect-label identity

Direct algebraic decomposition yields the imperfect-label identity. Set $(a, b, c) = (\tau_0, \tau_H, \tau_A)$ and $(u, m) = (u(s), \mu(s))$. Direct expansion gives

$$\begin{aligned}(a+b+c)(a+u^2b+m^2c) - (a+ub+mc)^2 \\ = ab(u-1)^2 + ac(m-1)^2 + bc(u-m)^2.\end{aligned}$$

Dividing by $(a+b+c)(a+ub+mc)^2$ yields (80). □

E.3 Additional comparative statics

The main text concentrates on the parameter groups that organize existence and type location. It also studies factual reliance. Table 4 varies the remaining user-side primitives one at a time. The exercise uses multipliers $\{0.5, 1, 1.5\}$. Parameter domains determine admissibility. The constraint $\delta_F \leq 1$ sets the high value at endpoint 1. The absolute-precision row varies T while holding $\tau/T = 2$. A mechanical adjustment of c_ℓ fixes audit slack. Every row preserves uniqueness and report monotonicity.

Table 4: Additional one-at-a-time comparative statics

	Primitive value			Model-implied harmed mass		
Primitive	Low	Middle	High	Low	Middle	High
A_H	0.5	1	1.5	0.450	0.458	0.461
B_A	0.175	0.35	0.525	0.462	0.458	0.454
P	0.5	1	1.5	0.146	0.458	0.518
ω	0.5	1	1.5	0.411	0.458	0.454
$c_v + c_p$	1.7	3.4	5.1	0.448	0.458	0.462
δ_F	0.35	0.70	1.00	0.524	0.458	0.286
T at fixed τ/T and audit slack	0.5	1	1.5	0.294	0.458	0.532

The centered normal type distribution has standard deviation 3. The total report-cost row holds $c_v/(c_v + c_p)$ fixed. The P rows vary a reduced-form loading. Equation (5) and Figure 1 report feasible joint movements in (τ, P).

The joint grid varies four dimensions simultaneously. Table 5 uses the Cartesian product

$$\{1.7, 3.4, 5.1\}_{c_v+c_p} \times \{0.35, 0.70, 1\}_{\delta_F} \times \{0.5, 1, 1.5\}_T \times \{0, 0.2, 0.5, 0.8, 1\}_s, \qquad \tau/T = 2.$$

This exercise holds $c_\ell = 6$ fixed, which allows audit slack to move. The 135 cells form a deterministic comparative-static grid. The table reports every provenance stratum alongside the pooled row over the declared support.

Table 5: Joint comparative statics by provenance

s	Median harm	Zero	Below 5%	Median cost diff.	AI cost lower
0	0.525	22.2%	22.2%	0.311	33.3%
0.2	0.458	22.2%	22.2%	0.124	40.7%
0.5	0.253	22.2%	22.2%	−0.194	51.9%
0.8	0.0046	22.2%	92.6%	−0.463	63.0%
1	0	100%	100%	−0.508	66.7%
All 135 cells	0.035	37.8%	51.9%	−0.044	51.1%

Each provenance row contains 27 cells. "Median full excess" denotes mean all-user full cost under AI minus no AI. Across all 135 cells, harmed mass ranges from 0 to 0.683. Mean full-cost excess ranges from −2.575 to 1.847.

E.4 Computational methods

The computation follows the analytical structure of the model. The unique reduced first-order condition determines every optimal report. The benchmark harm cutoffs use the analytical interval in (44). Gauss-Hermite integration evaluates smooth all-user moments. Optional-use welfare forms a lower envelope with kinks at the adoption boundaries. Adaptive quadrature on each smooth interval evaluates its expectation. Every comparative-static value re-solves the user problem before computing losses and probability masses.

AI Declaration

During the preparation of this work, the author used ChatGPT (OpenAI) solely for grammar checking, language polishing, and assistance with LaTeX formatting. The author reviewed and edited all AI-assisted output as needed and takes full responsibility for the content of the manuscript.

References

Itai Arieli, Omer Madmon, and Moshe Tennenholtz. Reputation-based persuasion platforms. *Games and Economic Behavior*, 147:128–147, 2024. doi: 10.1016/j.geb.2024.07.002.

Dirk Bergemann and Stephen Morris. Bayes correlated equilibrium and the comparison of information structures in games. *Theoretical Economics*, 11(2):487–522, 2016. doi: 10.3982/TE1808.

David Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.

Inácio Bó, Li Chen, and Rustamdjan Hakimov. Strategic responses to personalized pricing and demand for privacy: An experiment. *Games and Economic Behavior*, 148:487–516, 2024. doi: 10.1016/j.geb.2024.10.008.

J. Aislinn Bohren and Daniel N. Hauser. Misspecified models in learning and

games. *Annual Review of Economics*, 17:427–451, 2025. doi: 10.1146/annurev-economics-091324-032415.

Andrew Caplin and Mark Dean. Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*, 105(7):2183–2203, 2015. doi: 10.1257/aer.20140117.

Noah Castelo, Maarten W. Bos, and Donald R. Lehmann. Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5):809–825, 2019. doi: 10.1177/0022243719851788.

Alexander Clyde. Proxy variables and feedback effects in decision making. *Games and Economic Behavior*, 153:408–429, 2025. doi: 10.1016/j.geb.2025.07.008.

Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6):1431–1451, 1982. doi: 10.2307/1913390.

Peter M. DeMarzo, Dimitri Vayanos, and Jeffrey Zwiebel. Persuasion bias, social influence, and unidimensional opinions. *Quarterly Journal of Economics*, 118(3):909–968, 2003. doi: 10.1162/00335530360698469.

Berkeley J. Dietvorst, Joseph P. Simmons, and Cade Massey. Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1):114–126, 2015. doi: 10.1037/xge0000033.

Benjamin Enke and Florian Zimmermann. Correlation neglect in belief formation. *Review of Economic Studies*, 86(1):313–332, 2019. doi: 10.1093/restud/rdx081.

Ignacio Esponda and Demian Pouzo. Berk–nash equilibrium: A framework for modeling agents with misspecified models. *Econometrica*, 84(3):1093–1130, 2016. doi: 10.3982/ECTA12609.

Erik Eyster and Matthew Rabin. Cursed equilibrium. *Econometrica*, 73(5):1623–1672, 2005. doi: 10.1111/j.1468-0262.2005.00631.x.

Erik Eyster and Matthew Rabin. Naive herding in rich-information settings. *American Economic Journal: Microeconomics*, 2(4):221–243, 2010. doi: 10.1257/mic.2.4.221.

Ronald Fagin and Joseph Y. Halpern. Belief, awareness, and limited reasoning. *Artificial Intelligence*, 34(1):39–76, 1987. doi: 10.1016/0004-3702(87)90003-8.

Emel Filiz-Ozbay and Zhenxun Liu. Information manipulation of wishful thinkers. *Games and Economic Behavior*, 158:763–786, 2026. doi: 10.1016/j.geb.2026.05.010.

Mira Frick, Ryota Iijima, and Yuhta Ishii. Misinterpreting others and the fragility of social learning. *Econometrica*, 88(6):2281–2328, 2020. doi: 10.3982/ECTA16981.

Drew Fudenberg, Giacomo Lanzani, and Philipp Strack. Selective-memory equilibrium. *Journal of Political Economy*, 132(12):3978–4020, 2024. doi: 10.1086/731412.

Xavier Gabaix. A sparsity-based model of bounded rationality. *Quarterly Journal of Economics*, 129(4):1661–1710, 2014. doi: 10.1093/qje/qju024.

Nicola Gennaioli and Andrei Shleifer. What comes to mind. *Quarterly Journal of Economics*, 125(4):1399–1433, 2010. doi: 10.1162/qjec.2010.125.4.1399.

Tobias Gesche. De-biasing strategic communication. *Games and Economic Behavior*, 130:452–464, 2021. doi: 10.1016/j.geb.2021.09.001.

Ronen Gradwohl and Artyom Jelnov. Partial credence goods on review platforms. *Games and Economic Behavior*, 148:517–534, 2024. doi: 10.1016/j.geb.2024.11.002.

Ben Greiner, Philipp Grünwald, Thomas Lindner, Georg Lintner, and Martin Wiernsperger. Incentives, framing, and reliance on algorithmic advice: An experimental study. *Management Science*, 72(1):302–322, 2026. doi: 10.1287/mnsc.2022.02777.

Joseph Y. Halpern. Alternative semantics for unawareness. *Games and Economic Behavior*, 37(2):321–339, 2001. doi: 10.1006/game.2000.0832.

Joseph Y. Halpern and Willemien Kets. Ambiguous language and common priors. *Games and Economic Behavior*, 90:171–180, 2015. doi: 10.1016/j.geb.2015.01.007.

Aviad Heifetz, Martin Meier, and Burkhard C. Schipper. A canonical model for interactive unawareness. *Games and Economic Behavior*, 62(1):304–324, 2008. doi: 10.1016/j.geb.2007.07.003.

Shota Ichihashi and Alex Smolin. Data provision to an informed seller. *Games and Economic Behavior*, 153:131–144, 2025. doi: 10.1016/j.geb.2025.06.002.

Philippe Jehiel. Analogy-based expectation equilibrium. *Journal of Economic Theory*, 123(2):81–104, 2005. doi: 10.1016/j.jet.2003.12.003.

Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011. doi: 10.1257/aer.101.6.2590.

David Lagziel and Yevgeny Tsodikovich. Working with AI: An analysis for rational integration. *Games and Economic Behavior*, 153:254–267, 2025. doi: 10.1016/j.geb.2025.06.009.

Gilat Levy and Ronny Razin. Correlation neglect, voting behavior, and information aggregation. *American Economic Review*, 105(4):1634–1645, 2015. doi: 10.1257/aer.20140134.

Jing Li. Information structures with unawareness. *Journal of Economic Theory*, 144 (3):977–993, 2009. doi: 10.1016/j.jet.2008.10.001.

Jennifer M. Logg, Julia A. Minson, and Don A. Moore. Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151:90–103, 2019. doi: 10.1016/j.obhdp.2018.12.005.

Stefania Minardi and Andrei Savochkin. Time for memorable consumption. *Games and Economic Behavior*, 148:296–322, 2024. doi: 10.1016/j.geb.2024.09.010.

Sendhil Mullainathan. A memory-based model of bounded rationality. *Quarterly Journal of Economics*, 117(3):735–774, 2002. doi: 10.1162/003355302760193887.

Sendhil Mullainathan, Joshua Schwartzstein, and Andrei Shleifer. Coarse thinking and persuasion. *Quarterly Journal of Economics*, 123(2):577–619, 2008. doi: 10.1162/qjec.2008.123.2.577.

Paula Onuchic. Advisors with hidden motives. *Games and Economic Behavior*, 153: 431–450, 2025. doi: 10.1016/j.geb.2025.07.009.

Joshua Schwartzstein. Selective attention and learning. *Journal of the European Economic Association*, 12(6):1423–1452, 2014. doi: 10.1111/jeea.12104.

Ran Spiegler. "Data Monkeys": A procedural model of extrapolation from partial statistics. *Review of Economic Studies*, 84(4):1818–1841, 2017. doi: 10.1093/restud/rdx004.

Andrea Wilson. Bounded memory and biases in information processing. *Econometrica*, 82(6):2257–2294, 2014. doi: 10.3982/ECTA12188.